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Divide and Indebt

How Geoeconomic Fragmentation Affects Public Debt and Macro-Fiscal Stability

Prepared by Davide Furceri, Marcos Poplawski-Ribeiro, Ervin Prifti

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Divide and Indebt: How Geoeconomic Fragmentation Affects Public Debt and Macro-Fiscal Stability*

Prepared by Davide Furceri, Marcos Poplawski-Ribeiro, Ervin Prifti

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ABSTRACT: This paper quantifies the macro-fiscal consequences of geoeconomic fragmentation using the Geopolitical Fragmentation Index of Fernández Villaverde, Mineyama, and Song (2024). Using a panel of 90 economies over 1990–2022, we identify structural fragmentation shocks in a panel SVAR and trace their dynamic effects through local projections, allowing for heterogeneity across country groups. We find that a one standard deviation increase in geoeconomic fragmentation raises public debt by up to 3 percentage points of GDP in the medium term. The debt buildup reflects weaker economic activity, wider fiscal deficits driven by higher spending and lower revenues, and a temporary tightening of financing conditions. The effects are stronger in advanced economies and in countries more exposed to geopolitically distant trade partners. Geoeconomic fragmentation also amplifies uncertainty around future debt and growth and increases the risk of tail events.

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WORKING PAPERS

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I. Introduction

The recent surge in geoeconomic fragmentation—defined as the unraveling of global economic integration (Aiyar and others, 2023)—has reshaped international trade, capital flows, and macroeconomic policy. Rising protectionism, strategic decoupling, and the growing use of economic instruments to pursue geopolitical objectives reflect deeper national security concerns, ideological divides, and competition over critical technologies and resources. As countries pursue "friendshoring," restrict access to advanced technologies, and reconfigure supply chains along political lines, the global economy is undergoing a structural transformation with potentially important macro-fiscal consequences (Javorcik and others, 2024; Clayton, Maggiori and Schreger, 2025).

Since the COVID-19 pandemic, rising geopolitical tensions and economic fragmentation—as captured by the Geopolitical Fragmentation Index developed by Fernández Villaverde, Mineyama, and Song (2024)—have coincided with a sharp increase in global public debt from already elevated levels (Figure 1; IMF, 2026). This co-movement raises a central question: *does geoeconomic fragmentation causally contribute to rising public debt?*

This paper advances the literature by providing evidence on the impact of geoeconomic fragmentation on public debt and by quantifying the macro-fiscal channels driving debt dynamics. While a growing literature documents the real and financial effects of fragmentation (Aiyar, Presbitero, and Ruta, 2023), its fiscal implications remain largely unexplored. Existing work shows that fragmentation reduces trade, investment, and output, and increases financial volatility and sovereign risk *premia*. However, there is little causal evidence on how these effects translate into fiscal imbalances and public debt dynamics, constituting an important gap in the literature. If fragmentation systematically weakens growth while inducing fiscal expansion, it may represent a persistent source of debt accumulation, with direct implications for fiscal sustainability and macroeconomic stability. To the best of our knowledge, this is the first paper attempting to address this gap.

From a fiscal perspective, fragmentation operates through several mutually reinforcing channels. On the revenue side, protectionist measures may temporarily raise tariff income, but lower trade volumes, weaker competitiveness, and slower growth can ultimately narrow the tax base (Furceri and others, 2018; Barattieri, Cacciatore and Ghironi, 2021). On the expenditure side, governments may respond to trade disruptions and industrial dislocation with subsidies, compensation schemes, and higher public investment, thereby widening fiscal deficits. Technological decoupling in sectors such as semiconductors and artificial intelligence can further raise public spending as countries seek to replace foreign inputs with domestic capacity (Mano and others, 2021). Energy fragmentation adds further strain, as price volatility and supply risks prompt costly support programs while dampening economic activity (Gardes-Landolfini and others, 2023). More broadly, the resurgence of industrial policy to promote reshoring and strategic autonomy through tax credits, infrastructure spending, and sector-specific subsidies may increase deficits and debt (Antràs, 2024; Baquie and others, 2025). In parallel, financial fragmentation can tighten financing conditions and raise sovereign borrowing costs, especially in economies exposed to global spillovers (IMF, 2023a; IMF, 2025).

To quantify the effects of geoeconomic fragmentation on public debt, we adopt a two-stage empirical strategy. First, we identify structural geoeconomic fragmentation shocks in a panel structural vector autoregression (SVAR) using the Geopolitical Fragmentation Index (GFI) of Fernández Villaverde, Mineyama, and Song (2024).[1] Within this internally consistent framework, we consider two alternative identification

[1] Relative to the April 2025 Fiscal Monitor where part of this analysis was initially featured, we switch to a baseline identification strategy that relies on a SVAR with recursive restrictions estimated on a pooled panel of countries, whereas the Fiscal Monitor

strategies: one based on recursive short-run restrictions and another based on the external narrative instrument used by Fernández Villaverde, Mineyama, and Song (2024). Second, we use the identified shocks in a local projection framework to study heterogeneous effects across countries. The results, based on an unbalanced panel of 90 economies over 1990–2022, indicate that a one-standard-deviation increase in geoeconomic fragmentation raises public debt persistently by about 1 to 3 percentage points of GDP in the medium term.

This effect operates through a set of macro-fiscal transmission channels. Following a fragmentation shock, real output falls by about 1.5 to 1.6 percent on impact and remains persistently below trend, temporarily eroding revenues, while persistently higher spending—by about 0.4 percentage point of GDP in the medium run—widens the fiscal deficit. At the same time, sovereign spreads increase temporarily, by roughly 15 basis points, reinforcing short-run debt accumulation.

These average effects mask important heterogeneity across country groups and help clarify the mechanisms at work. Using state-dependent local projections with smooth-transition interactions, we find that advanced and Western economies experience larger short-run increases in public debt—about 1 to 2 percentage points of GDP more than other economies—because fragmentation shocks generate both stronger real disruptions and more forceful fiscal responses. In these economies, deeper trade and financial integration make production networks more vulnerable to disruptions in global linkages, while larger automatic stabilizers and greater fiscal space allow governments to cushion households and firms through higher spending. Countries more exposed to geopolitically distant trade partners also experience larger increases in debt, reflecting greater vulnerability to supply-chain disruptions and input shortages when fragmentation rises. Weaker activity then erodes revenues and prompts fiscal support, leading to larger deficits and debt accumulation. Finally, fragmentation amplifies macro-financial risks beyond its effects on average debt and growth. It raises debt-at-risk and worsens growth-at-risk by increasing uncertainty around future outcomes, shifting the upper tail of the debt distribution upward and the lower tail of the growth distribution downward (Adrian, Boyarchenko, and Giannone, 2019; Furceri and others, 2025). Fragmentation therefore emerges not only as a source of higher average debt, but also as a structural macro-fiscal risk that increases the likelihood of adverse fiscal and macroeconomic tail events.

These findings relate to three strands of the literature. First, a growing body of work documents the real effects of de-globalization and “friendshoring,” showing that fragmentation reduces trade, weakens productivity, and lowers output (Constantinescu, Mattoo and Ruta, 2020; Aiyar, Presbitero and Ruta, 2023; Attinasi, Ciccarelli and De Santis, 2023; Campos and others, 2023; and IMF, 2023b). We contribute to this strand by showing that these real effects have material fiscal implications, as output losses and disrupted production networks translate into weaker revenues, larger deficits, and higher public debt. Second, a related literature studies financial channels, finding that geopolitical risk raises sovereign risk premia, disrupts capital flows, and increases financial volatility (IMF, 2023a, 2023c; Afonso, Alves, and Monteiro, 2024, 2025; Choi and Havel, 2024; Huang and others, 2024; and Jiang and others, 2024). We add to this literature by showing that the fiscal consequences of fragmentation are not driven primarily by persistent increases in sovereign borrowing costs, documenting only a temporary financial tightening. Third, recent work has developed new measures of fragmentation and examined their macroeconomic effects, notably Fernández Villaverde, Mineyama, and Song (2024), who document negative output effects. We build on this measurement and identification agenda by tracing the transmission of geoeconomic fragmentation shocks to public finances and by documenting their effects not only on average debt dynamics, but also on debt-at-risk and growth-at-risk.

analysis relied on Local Projections. We also expand the country coverage, introduce additional dimensions of heterogeneity including geopolitical trade exposure. The location -scale model was already included in the Fiscal Monitor.

The remainder of the paper is structured as follows. Section II estimates the baseline effects on public debt and the main macro-fiscal transmission channels for that effect. Section III examines heterogeneity across country groups, outlining the empirical strategy and documenting differential effects. Section IV analyzes the implications of fragmentation for the medium-term distributions of future public debt and growth. Section V concludes.

II. Effects on Public Debt and Transmission Channels

This section analyzes how geoeconomic fragmentation affects public debt. To do so, we use the Geopolitical Fragmentation Index (GFI) introduced by Fernández Villaverde, Mineyama, and Song (2024), which provides a synthetic measure of the rise in global fragmentation by combining information from four domains: trade, finance, policy, and mobility. We first describe the data used in the analysis, then present the empirical methodology and baseline findings, and conclude with robustness checks of the main results.

A. Data and Sample

The GFI index is constructed using a dynamic-factor approach applied to a wide range of indicators that capture the dynamics of different manifestations of geopolitical and geoeconomic fragmentation, such as trade restrictions, capital-flow measures, mobility barriers, armed conflict intensity, and patterns of alignment in UN General Assembly voting. It is more comprehensive than other indexes—such as Caldara and Iacoviello's (2022) geopolitical risk index, which focuses on the risk associated with political instability and turmoil—as it incorporates policy measures, giving it a distinct geoeconomic flavor by highlighting how economic factors intertwine with geopolitical issues.[2] For instance, in the trade and investment sphere, the post-2008 slowdown in trade openness and the deceleration in cross-border capital flows signal a weakening of global integration. An additional example is the steady increase in sanctions and other extraterritorial measures on the trade policy side recently, which suggests more adversarial geopolitical relations across countries. In the mobility domain, in turn, heightened concerns over migration is one example which points to rising frictions in people's movement. Together, these trends have produced a broad and persistent retreat from global economic interconnectedness.

The macro-fiscal variables are drawn primarily from the IMF World Economic Outlook database. Country-specific macroeconomic variables include real GDP and inflation, with the latter measured as the log change in the consumer price index. Fiscal variables include the overall balance, total public expenditure, and total revenues, all expressed as shares of GDP, as well as sovereign spreads measured by long-term government bond yields relative to U.S. Treasuries. Global financial variables include the Chicago Fed National Financial Conditions Index, crude oil prices (WTI), the S&P 500 index, and the VIX, all retrieved from FRED.

Combining these variables yields an unbalanced panel of 90 countries over 1990–2022, comprising 2,515 country-year observations in the baseline estimation sample. The Appendix provides additional information on the database, including descriptive statistics (Table A1) and the list of countries in the estimation sample by income level (Table A2).

Figure 1 shows a clear positive relationship between public debt and geoeconomic fragmentation over most of the sample period. Geoeconomic fragmentation and average global indebtedness tend to rise and fall together, suggesting a growing alignment between geopolitical-economic tensions and fiscal dynamics. The comovement becomes more pronounced after the Global Financial Crisis, pointing to a stronger interaction

[2] While geopolitics concerns mainly the influence of geographic space, power rivalries, and political strategy in shaping global affairs and emphasizes military alliances, territorial control, and diplomatic maneuvering; the concept of geoeconomics captures the use of economic instruments—such as trade policy, investment restrictions, and financial leverage—as tools to achieve strategic geopolitical objectives.

between global economic dislocation and sovereign fiscal trajectories. The correlation appears particularly strong after 2016, coinciding with the acceleration of strategic competition, most notably between the United States and China, and the broader slowdown in trade (Aslam and others, 2018) and globalization (Aiyar and Ohnsorge, 2024).

Although the comovement between the Geopolitical Fragmentation Index (GFI) and global debt is strong, it may partly reflect spurious correlation. Both series trend upward and respond to common global shocks—such as financial crises—that can simultaneously raise geopolitical tensions and public debt. Feedback effects may also be present: fragmentation can weaken growth and increase debt ratios, while fiscal strain may in turn intensify political frictions. Moreover, global aggregates mask substantial cross-country heterogeneity. Without a structural identification strategy, the direction of causality therefore remains indeterminate. To address these concerns, we use both recursive identification and external instruments as the core identification strategies in our SVAR framework, as discussed in the next section.

B. Empirical Strategy

Our empirical strategy follows closely Fernández Villaverde, Mineyama, and Song (2024), based on a two-stage approach. In the first stage, we recover structural innovations to the Geopolitical Fragmentation Index (GFI) from a panel SVAR, using complementary identification schemes designed to isolate exogenous variations in fragmentation from concurrent global shocks and confounding factors. In the second stage, we treat these identified shocks as external impulses in a local projection (LP) framework (Jordà, 2005). This approach combines structural identification with the flexibility of LPs, delivering transparent estimates of the dynamic transmission from fragmentation to public finances. The LP framework further enables a systematic analysis of heterogeneity across country characteristics and provides a natural setting to compute debt-at-risk measures.

B.1. Panel SVAR identification

The reduced form of the VAR is:

$$\boldsymbol{Y}_{i,t} = \boldsymbol{\mu}_i + \sum_{j=1}^{p} \boldsymbol{\varphi}_j \boldsymbol{Y}_{i,t-\mathrm{j}} + \boldsymbol{u}_{i,t}, \qquad (1)$$

where $\boldsymbol{Y}_{i,t} = \{GFI_t, d_{i,t}, gdp_{i,t}, p_{i,t}, rev_{i,t}, exp_{i,t}, s_{i,t}, global\ variables\}$ includes 13 variables given by the geoeconomic fragmentation index, debt to GDP, log of real output, log of the CPI, public revenues and spending as a share of GDP, the spread on long term sovereign bond yields, all for country *i* in year t. To mitigate omitted variable bias arising from global conditions that could jointly affect the GFI and domestic macro-fiscal variables, the specification also includes key global controls: the log of global GDP; the log of spot crude oil prices (West Texas Intermediate); log of the S&P500 stock index; the VIX index; National Financial Conditions Index (NFCI), which provides U.S. financial conditions in money, debt and equity markets; and the yield on two year treasuries. The lag length of the VAR is set to $p = 2$. Country-specific intercepts ($\boldsymbol{\gamma}_i$) capture unobserved heterogeneity while the slope matrices ($\boldsymbol{\varphi}_j$) are common across countries.

The corresponding structural form of the VAR can be written as:

$$\boldsymbol{A}_0 \boldsymbol{Y}_{i,t} = \boldsymbol{\gamma}_i + \sum_{j=1}^{p} \boldsymbol{A}_j \boldsymbol{Y}_{i,t-j} + \boldsymbol{e}_{i,t}, \qquad (2)$$

where $\boldsymbol{\varphi}_j = \boldsymbol{A}_0^{-1} \boldsymbol{A}_j$ and $\boldsymbol{A}_0$ is the matrix of contemporaneous relationships between the variables in $\boldsymbol{Y}_{i,t}$ and the reduced form residuals related to the structural shocks through $\boldsymbol{u}_{i,t} = \boldsymbol{A}_0^{-1} \boldsymbol{e}_{i,t}$.

From the latter relationship one can derive the system of equations relating the unknown structural parameters to the known elements recovered from the estimation of the reduced form, as $\boldsymbol{B'}_0\boldsymbol{B}_0 = \boldsymbol{\Sigma}$, where the right-hand side is the covariance matrix of the reduced form residuals and $\boldsymbol{B}_0 = \boldsymbol{A}_0^{-1}$. Since there are more unknowns (k^2 elements of $\boldsymbol{B}_0$) than knowns ($\frac{k(k+1)}{2}$ elements of the symmetric matrix $\boldsymbol{\Sigma}$), restrictions must be imposed on the elements of $\boldsymbol{B}_0$ to exactly identify the system.

We employ two identification strategies to identify the structural shocks of geoeconomic fragmentation: one based on the recursive method; and another based on the same external narrative instrument used by Fernández Villaverde, Mineyama, and Song (2024). Our baseline identification strategy relies on a set of economically grounded short-run zero restrictions, which make $\boldsymbol{B}_0$ lower-triangular. In our baseline specification, we order the GFI first, assuming that it does not respond within the year to any of the other endogenous variables in $\boldsymbol{Y}_{i,t}$. Assuming short-term exogeneity of geoeconomic fragmentation is realistic for the country-level macro-fiscal block, as only a few countries have the power to affect global geopolitics through fiscal levers within the year, while the rest experience geopolitical developments as exogenous shocks or have negligible influence on global political alignments, at best. Moreover, countries often use non-fiscal instruments — such as political and diplomatic efforts, cultural diplomacy, ideological influence, and regulatory actions — to shape global alignments without significant fiscal implications.

For global variables, contemporaneous reverse feedback is more plausible. We therefore re-estimate the SVAR under an alternative ordering in which the GFI is ordered last. Even in this specification, however, it remains plausible that the GFI responds only sluggishly to short-term global financial shocks, since fragmentation reflects cumulative political-economic shifts rather than high-frequency market dynamics. As an intermediate case, we order the global block first, followed by the GFI and then the domestic macro-fiscal block. Under this ordering, the domestic block is assumed not to contemporaneously affect any preceding global variables, including the GFI, while the GFI is allowed to respond contemporaneously to other global shocks.

For our IV-SVAR estimator we use an external instrument Z_t built by Fernández Villaverde, Mineyama, and Song (2024) which takes the value of 1 in quarters of heightened geoeconomic fragmentation, -1 in quarters that signal opposite shocks pushing the world to greater integration, and zero otherwise. The full list of events by date is included in the Appendix (Table A3) and it includes dates in which the world moved towards higher integration, like China's admission to the WTO, or towards higher fragmentation, like NATO's intervention in Serbia. We aggregate the index to the annual level. The instrument is assumed to be correlated with the GFI shock but orthogonal to other shocks — i.e., $E\left[Z_t e_{i,GFI,t}\right] \neq 0$ and $E\left[Z_t e_{i,j,t}\right] = 0$ (for country i and $j \neq GFI$). Estimation follows the approach of Plagborg-Møller and Wolf (2021), whereby the instrument is added as the first variable to our baseline VAR. Using recursive identification implies that the impulse responses of the first shock are equivalent to the ones generated by the standard proxy-SVAR (Mertens and Ravn, 2013).

B.2. Estimation Using the Local Projection Method

As the next section discusses, the analysis of heterogeneous effects depending on countries' characteristics will rely on a linear Local Projection estimator. Hence, as for the baseline estimation mentioned, above, we also estimate the dynamic causal effects of geoeconomic fragmentation using the following linear LP specification:

$$y_{i,t+h} = \gamma_i^h + \sum_{j=0}^{p} b_j^h shock_{t-j} + \sum_{j=1}^{p} \boldsymbol{c}_j^h \boldsymbol{Y}_{i,t-j} + \varepsilon_{i,t+h}, \qquad (3)$$

where the coefficient b_0^h traces out the impulse response function (IRF) at horizon *h*; $shock_t$ is the previously identified structural shock common to all countries (e_t^{GFI}); and γ_i^h are country specific intercepts. The shock is assumed to be exogenous and affects each of the macro-fiscal variables without any contemporaneous feedback.

C. Results

C.1. Effects on Public Debt

Figure 2 reports impulse responses of the debt-to-GDP ratio to a one standard deviation innovation in geoeconomic fragmentation estimated under five alternative identification schemes. The top-left panel shows the recursive SVAR under the baseline ordering, the top-middle panel reports the corresponding local projection (LP) estimates based on Equation (3), the top-right reports a recursive model with the GFI ordered last. The bottom row presents an intermediate case in which the global block precedes the GFI and the macro-fiscal variables in the same order as in the previous case, and an instrumental variable SVAR.

Across specifications, fragmentation shocks are followed by a statistically significant and persistent increase in public debt, although the magnitude and short-run dynamics vary across identification schemes. Under the baseline recursive ordering, debt rises by slightly more than 3 percentage points within two years and then converges to a plateau of about 3 percentage points of GDP. The LP estimates (top-center panel) display a similar short-run response but imply a more muted medium-run effect, suggesting some attenuation once fewer dynamic restrictions are imposed. When the GFI is ordered last (top-right panel), the response is delayed and becomes significant only after several periods, consistent with a more restrictive contemporaneous structure; the medium-run effect remains positive but smaller, leveling off at around 0.5 percentage point of GDP. A similar pattern emerges when the global block is ordered first (bottom-left panel), although in this case the response is significant on impact and stabilizes at around 1 percentage point of GDP. Finally, the IV-SVAR estimates (bottom-center panel) lie between the baseline recursive and LP responses, with debt peaking after one year and settling at approximately 2 percentage points of GDP. To assess instrument relevance, we recover the reduced-form innovations from the instrument and GFI equations and regress the latter on the former. This first-stage analogue captures the contemporaneous link between unexpected narrative events and innovations to fragmentation. The coefficient is highly statistically significant, with a robust F-statistic of 1,031 and an R^2 of 0.24, indicating that narrative shocks explain about one-quarter of the variation in GFI innovations and provide a strong instrument.

To gauge the economic magnitude of the fragmentation shock, we compare its effect on public debt with the debt responses to selected country-specific and global innovations identified within the same baseline SVAR (Figure 3). This ensures that all responses are estimated using a common sample, information set, lag structure, identification scheme, and one-standard-deviation normalization. We consider shocks to real output, sovereign financing conditions, the fiscal balance, and selected global variables. Among these, the fragmentation shock generates the largest positive and most persistent increase in public debt, exceeding the effects of sizable fiscal and commodity-price innovations (i.e., the crude oil price WTI index), and displaying substantially greater persistence than the financial shocks (i.e., the sovereign bond spread and Chicago Fed National Financial Conditions Index—NCFI). These comparisons are intended as a within-model benchmark of economic magnitude rather than as exact counterparts to canonical output-gap, interest-growth differential, financial-crisis, or commodity-price shocks, since the corresponding VAR innovations may capture broader disturbances. The estimated effect is economically meaningful also relative to normal debt dynamics in the sample. Average public debt in the estimation sample is around 56 percent of GDP (see Table A.1 in the Appendix), so a 1–3 percentage-point increase raises the debt ratio by roughly 2–5 percent relative to its

average level. Put differently, the response corresponds to about 10–30 percent of the standard deviation of annual debt changes, which is 9.9 percentage points of GDP in the sample. While this is smaller than the debt movements observed during major crises, it is sizable for a single geoeconomic shock, especially because the effect is persistent over the medium term. Differences in timing and magnitude can be traced to the extent to which each scheme restricts contemporaneous interactions and isolates exogenous variation in fragmentation. Our preferred specification is the baseline recursive SVAR (top-left chart in Figure 2), which delivers an IRF identification that is transparent and with a theory-consistent ordering. That specification also avoids reliance on external instruments, whose validity is difficult to verify and whose finite-sample performance may be compromised when relevance is limited.

Finally, to examine which dimensions of fragmentation matter most for debt dynamics, we re-estimate the baseline SVAR by replacing the aggregate GFI, one at a time, with each of its four subcomponents: trade, financial, mobility, and political fragmentation. Shocks to each subcomponent generate a positive and statistically significant increase in public debt. The effects are largest for the trade and political components, with debt rising by about 2 percentage points on impact and peaking at close to 3 percentage points after one year (Figure 4). Financial and mobility fragmentation generate somewhat smaller initial effects, but their debt responses subsequently increase to about 2–2.3 percentage points. Overall, these results closely mirror the magnitude and persistence of the baseline estimates, indicating that the debt effect reflects several dimensions of fragmentation rather than any single conflict-heavy component.

C.2. Robustness checks

This subsection presents the results of several sensitivity checks to our baseline SVAR estimate to assess the robustness of our results and identification strategy. Our first robustness check estimates a sign-restricted Bayesian SVAR following the approach of Arias, Rubio-Ramírez and Waggoner (2018). We impose sign restrictions on the contemporaneous impulse responses to a geoeconomic fragmentation shock that reflect theoretically plausible short-run effects. Specifically, a positive GFI shock (higher fragmentation) is assumed to increase the fragmentation index, sovereign spreads and public debt, while reducing real output and the fiscal balance. To improve numerical stability and rule out economically implausible rotations, we also impose minimal restrictions on fiscal policy shocks, assuming that a fiscal tightening shock reduces output, improves the fiscal balance, and lowers public debt. We adopt a Jeffreys (uninformative) prior, allowing the likelihood—together with the identifying sign restrictions—to determine the set of admissible structural models and the associated posterior distribution of impulse responses (Ferroni and Canova, 2021). Because the identifying restrictions are imposed only on the contemporaneous responses, the positive impact responses of debt and the negative impact response of the fiscal balance are part of the identification strategy. By contrast, the medium-run persistence and magnitude of the impulse responses are determined by the estimated VAR dynamics and therefore provide an informative robustness check relative to the baseline specification.

Table 1 shows the sign restrictions of each shock (rows) on a given variable (columns). To account for cross-country heterogeneity, we relax the assumption of common slopes and estimate the SVARs separately for each country and compute average impulse responses using the mean group (MG) approach of Pesaran and Smith (1995). This produces more robust and representative estimates in the presence of cross-country heterogeneity but also makes the issue of over-parametrization in classical VARs more acute due to the reduced sample size at the individual country level. Hence, wider confidence bands are expected. We then estimate the reduced form VAR and obtain a factorization of the covariance matrix $\boldsymbol{\Sigma} = \boldsymbol{B}_0{}'\boldsymbol{B}_0$ as $\boldsymbol{P}'\boldsymbol{P}$ (e.g., via Cholesky decomposition). That matrix allows us to generate a large number of candidate impact matrices as

$\boldsymbol{B}_0 = \boldsymbol{P}\,\boldsymbol{Q}$, where $\boldsymbol{Q}$ is a random orthonormal matrix drawn from the Haar distribution, from which only those $\boldsymbol{Q}$ matrices whose implied IRFs satisfy the sign restrictions are retained.[3]

Figure 5 displays the results of these three robustness checks. The first robustness check (Figure 5, left panel) shows IRFs converging to about 2 percentage points of GDP in the medium term, closely mirroring the baseline pooled estimate. Such results indicate that the pooled specification is not materially biased by slope homogeneity assumptions and that the debt response reflects a common transmission mechanism across countries. Importantly, the alignment of the IRFs across identification schemes suggests that the result is not driven by a particular identification strategy. Overall, the similarity of the mean-group and pooled average responses suggests that the baseline average effect is not driven by the common-slope restriction.

To ensure that our results are not driven by large geopolitically pivotal economies, our second robustness check re-estimates the baseline SVAR after excluding the United States and China. Both economies play a central role in generating fragmentation shocks and their fiscal-policy reactions to geopolitical developments, also given the size of their economies, can mechanically feed into the GFI. Leaving them out helps us verify whether the estimated responses are not disproportionately shaped by idiosyncratic dynamics in the world's two largest economies. Impulse responses from our second robustness check excluding the United States and China are moreover similar to the baseline specification using the full sample (Figure 5, middle panel). This result demonstrates that the effects of fragmentation on public debt are not driven by developments that pertain exclusively to the U.S. and China. That reinforces the validity of the assumptions of the baseline specification with GFI ordered first, requiring no effects within the year from the fiscal block and the global variables on GFI itself (Figure 5, middle panel).

Our third robustness check implements a placebo test based on a permuted instrument that randomly reassigns the timing of fragmentation and integration events across the sample while preserving the empirical distribution of event indicators. This procedure breaks the link between actual event dates and macroeconomic conditions but retains the overall frequency of events. Re-estimating the IV-SVAR with this placebo instrument allows us to test whether the baseline responses are driven by the true timing of geopolitical shocks rather than by incidental correlations. If the placebo instrument yields impulse responses that are centered around zero, this provides evidence against spurious timing-based correlations as an explanation for our baseline results (Burchardi and others, 2020; Konrad, Schularick and von Hagen, 2020; Rose and Shem-Tov, 2023).[4] Results from the third robustness check also validate our IV-SVAR and baseline estimation (Figure 5, right panel). When using a placebo instrumental variable with randomly permuted values, the estimated IRFs obtained from 300 permutations average closely to zero, providing suggestive evidence on the validity of the original instrument.

As a further robustness check of the external-instrument identification, we address concerns that some narrative events may affect fiscal outcomes directly rather than exclusively through geoeconomic fragmentation. We therefore re-estimate the IV-SVAR after excluding the events with the clearest potential for broad global fiscal or macroeconomic effects: the Iraqi invasion of Kuwait (1990Q3), the Gulf War (1991Q1), the September 11 attacks (2001Q3), and Russia's invasion of Ukraine (2022Q1). These episodes are particularly likely to have affected fiscal outcomes through channels such as oil prices, defense and security spending, risk aversion, and emergency policy support. The resulting debt IRF remains broadly consistent with the baseline IV-SVAR estimates – although not in precise timing and magnitude – suggesting that the

[3] Although all matrices are country-specific in this case, we have omitted the *i* subscript in this paragraph for simplification.

[4] We have also performed additional robustness checks using United Nations Security Council cases and decisions as instruments for GFI. Although the results were qualitatively similar, the first-stage of the estimations and the exclusion restriction tests were not strong enough to support their usage as instrument (not shown here, but available upon request).

estimated responses are not driven by the small number of military and terrorist events that may plausibly have direct fiscal effects (Figure 6).

As an additional robustness check, we explicitly account for major global crisis episodes and alternative measures of geopolitical risk. We augment the baseline SVAR with dummy variables for the Global Financial Crisis and the COVID-19 pandemic and the country-year Caldara-Iacoviello Geopolitical Risk Index, which captures a narrower dimension of geopolitical tensions related to conflicts, war threats, and terrorism and is one component of the broader fragmentation index of Fernández Villaverde, Mineyama, and Song (2024). The resulting debt IRF remains very close to the baseline estimate, indicating that the main findings are robust to controlling for major global crisis episodes and for a narrower measure of geopolitical risk (Figure 7).

Overall, these results support the validity of our findings across identification schemes and specifications, while establishing a plausible range for the estimated effects of geoeconomic fragmentation on public debt.

C.3. Macroeconomic Transmission Channels

Next, we turn to the impulse responses of the macro-fiscal block in the baseline SVAR (Figure 8) to shed light on the transmission channels through which geoeconomic fragmentation affects public debt. The dynamics can be interpreted through the standard debt accumulation identity, which links the evolution of the debt-to-GDP ratio to the fiscal balance, real activity, inflation, and the interest-growth differential. Taken together, the responses point to a set of mutually reinforcing general equilibrium mechanism.

Fragmentation shocks combine supply-side disruptions—such as trade interruptions, reduced efficiency of global value chains, and higher production costs associated with reshoring—with demand-side effects operating through heightened uncertainty and financial risks. The results suggest that supply-side forces dominate: real output falls sharply on impact, by about 1.6 percent, and remains persistently below trend; while the CPI rises with a lag, reaching about 0.7 percent in the medium run, consistent with cost-push pressures as firms gradually pass higher input and trade costs on to consumers.

The contraction in output raises the debt-to-GDP ratio mechanically by lowering its denominator, while also weakening fiscal revenues through a narrower tax base. The observed decline in revenues is consistent with this channel, particularly in economies with high trade openness and cyclically sensitive tax systems. The fiscal response reflects both automatic stabilizers and discretionary policy measures. The overall deficit widens by about 1 percentage point of GDP on impact and remains persistently elevated. Decomposing this response shows that public expenditure rises significantly—by around 0.9 percentage point of GDP on impact—likely reflecting a combination of countercyclical support, higher social spending, and structurally higher outlays related to defense, industrial policy, and supply-chain reconfiguration. Over time, expenditure partially retraces, suggesting fiscal consolidation efforts or the unwinding of temporary measures. Revenues, by contrast, fall on impact by about 0.1 percentage point of GDP and recover only gradually, reinforcing the initial widening of the deficit. The decomposition of revenues and spending shows that the deficit increase is driven mainly by lower indirect revenues and higher current expenditures (Figure 9).[5] Following a fragmentation shock, indirect taxes decline persistently, consistent with weaker consumption and trade activity, while direct taxes show little systematic response apart from a delayed increase after three years, possibly reflecting progressive tax systems or bracket-creep effects from higher inflation (Balasundharam, Kayastha and Poplawski-Ribeiro, 2023). On the spending side, current expenditures rise significantly on impact and remain elevated, reflecting automatic stabilizers, discretionary support to households and firms, possibly higher interest payments, and higher government consumption, including defense spending (Furceri and others, 2026). By contrast, capital

[5] The analysis is carried out using the SVAR with recursive identification by replacing the total spending and revenues with their subcomponents.

spending remains broadly flat, suggesting that governments prioritize short-term support over investment after fragmentation shocks.

Fragmentation also affects debt dynamics through financial markets. A one-standard-deviation shock raises sovereign spreads by about 15 basis points on impact, signaling higher risk premia amid greater global risk aversion, flight-to-safety dynamics, and uncertainty about fiscal and macroeconomic fundamentals. Although spreads measure borrowing costs relative to benchmark safe rates rather than absolute yields, a sustained widening would increase the marginal cost of sovereign borrowing.

These average responses may mask substantial heterogeneity across countries, depending on their fiscal space, exposure to global trade and finance, and geopolitical position. We therefore turn to the heterogeneity analysis in the next section.

III. Heterogeneity Analysis

Using the data and empirical strategy described above, this section estimates the heterogeneous effects of geoeconomic fragmentation on public debt across countries. Building on the methodology discussed in Subsection II.B.2, the first section presents the empirical strategy for these heterogeneity analyses. The second subsection reports the estimation results for the different types of country grouping.

A. Empirical Strategy

To analyze the transmission channels of geoeconomic fragmentation shocks and assess heterogeneity across country characteristics, we embed the structurally identified shocks from the panel SVAR into a panel local projection (LP) framework. This approach preserves the credibility of the shock identification while allowing dynamic responses to vary flexibly across outcome variables and state-dependent dimensions.

To study heterogeneity across observed characteristics, we augment the baseline local projection with smooth-transition interactions between the common fragmentation shock and a country-specific logistic transition function, to obtain a form of state-dependent effects:

$$y_{i,t+h} = \gamma_i + \tau_t + \sum_{j=1}^{p} \boldsymbol{c}_j^h \boldsymbol{Y}_{i,t-j} + \ldots$$

$$+ \; S(m,\theta)\left(\sum_{j=0}^{p} b_j^{h,s} shock_{t-j} + \sum_{j=1}^{p} \boldsymbol{c}_j^{h,s} \boldsymbol{Y}_{i,t-j}\right) + \; \varepsilon_{i,t+h}, \qquad (4)$$

where $S(m,\theta)$ is a logistic function $\frac{e^{(\theta m)}}{1+e^{(\theta m)}}$; m is usually a recentered and rescaled version of the original state variable; and θ governs the steepness of transition across groups and can be tuned to reflect the degree of smoothness in the shift between regimes. Because the fragmentation shock is common across countries, its non-interacted component is absorbed by the year fixed effects. Consequently, the interaction term identifies differential responses across countries as a function of the state variable, while the average response is captured by the time effects.

This state-dependent specification avoids the limitations of interactions based on group dummies, such as high- versus low-income classifications, which impose discrete sample splits and abrupt changes in marginal effects across predefined categories (Auerbach and Gorodnichenko, 2013; Eminidou, Geiger, and Zachariadis, 2023). Because the shock is interacted with a variable that varies across both countries and time, we include

year fixed effects to absorb common global factors that may jointly affect macro-fiscal outcomes and the GFI, thereby strengthening identification.

We examine several dimensions of heterogeneity, corresponding to alternative specifications of the variable m in Equation (4): (i) *income level*, for which m is the GDP per capita recentered at the World Bank's high-income threshold and scaled by its standard deviation; (ii) *trade exposure to geopolitically distant partners*, for which m is the weighted average import dependence on each country's trading partners with weights given by the bilateral absolute differences in Ideal Point Distance (IPD) in UN General Assembly voting, already standardized, capturing alignment with major geopolitical blocs; and (iii) *geopolitical alignment*, for which m is the country's IPD (Bailey, Strezhnev, and Voeten, 2017), expressed in standardized form; (iv) *structural characteristics*, such as institutional quality and capital account openness. The state variables entering the transition function is lagged by one year to reduce potential for contemporaneous feedback and endogeneity.

The parameter θ varies depending on the specific variable m examined to produce a moderate degree of smoothness, though estimates remain robust to a wide range of values. Since θ is positive, the transition function is close to zero for low levels of m. In contrast, for high levels of m, $S(m, \theta)$ is close to 1. Hence, $b_0^{h,s}$ identifies the cross-group differential IRF between a high- and low-state regime.

B. Results

Building on the empirical strategy outlined above, this section presents the results for each of the five heterogeneity dimensions. For each dimension, we report both the cross-group differential effects of geoeconomic fragmentation on public debt across regimes and the corresponding differential responses of macro-fiscal variables that capture the underlying transmission channels.

B.1. High- vs. Low- and Middle-Income Economies

In this section, we examine how the effects of geoeconomic fragmentation vary with a country's income level, using GDP per capita as the state variable. The differential response of public debt between high-income economies and low- and middle-income economies is sizable in the short run (1.2 percentage point of GDP) and fades thereafter (Figure 10).

The decomposition of debt dynamics sheds light on the underlying mechanisms (Figure 11). Fragmentation leads to a more pronounced and persistent output contraction in high-income countries, about 1 percent larger than in low- and middle-income countries. Because the smooth-transition specification identifies differential effects, this should be interpreted as a larger relative decline in output among high-income countries. On the fiscal side, advanced economies display a stronger spending response, while revenue dynamics remain broadly similar across groups. This pattern is consistent with greater fiscal space and stronger automatic stabilizers, which allow high-income economies governments to cushion adverse shocks more forcefully. Consequently, fiscal balances deteriorate more in advanced economies, contributing to the larger short-run increase in debt. By contrast, sovereign spreads show no statistically significant differential response across income groups, suggesting that global financial factors dominate the pricing of sovereign risk.

B.2. Trade Exposure to Geopolitically Distant Partners

Next, we investigate how trade exposure to potential geoeconomic opponents affects overall public debt and external debt. To do so, we build an index of Import Dependence on Geopolitical Distance (IDGDI), which measures the extent to which a country relies on imports from trade partners that are ideologically distant from itself. Ideological distance is measured using the country's Ideal Point Distance (IPD) – the difference between the underlying ideal point score of each country. The ideal point score corresponds to numerical estimates representing a country's foreign policy preferences along a latent ideological dimension, derived from its voting

behavior in the United Nations General Assembly. The underlying dimension reflects alignment with the U.S.-led international order, with higher IP scores indicating closer alignment with the West. The IDGDI in turn, is constructed by calculating the absolute difference in ideal point scores between the importing country and each of its trade partners—capturing the geopolitical distance—and then weighing these distances by the share of imports each partner contributes to the country's total imports. A higher IDGDI indicates greater dependence on ideologically distant countries, suggesting increased exposure to the weaponization of supply chains by geopolitical rivals. The import-share and the ideal point estimates are taken at t-1, prior to the realization of the fragmentation shock.

The positive differential IRF indicates that public debt increases more in countries with higher trade exposure to geopolitically distant partners relative to those with lower exposure (Figure 12). This pattern can be understood through the interaction of trade disruptions and fiscal responses (Figure 13). First, trade-related mechanisms play a central role. High import dependence on geopolitically distant partners makes countries more vulnerable to supply-chain disruptions when fragmentation intensifies. These economies face shortages of critical inputs, delays, and heightened uncertainty about future access to foreign suppliers. As a result, production is disrupted, investment is postponed, and output declines more sharply relative to less exposed countries. Second, these real-side effects translate into fiscal pressures. Weaker activity erodes tax revenues, while governments respond with increased spending to support households and firms and to facilitate supply-chain adjustment. This leads to a larger deterioration in fiscal balances in more exposed economies.

Importantly, we do not find evidence that these pressures are offset by tighter financing constraints. Sovereign spreads do not increase differentially across groups, suggesting that financial markets do not penalize high-exposure countries with higher borrowing costs. Consistent with this, we also do not observe a differential adjustment in external debt, indicating that fragmentation does not trigger precautionary deleveraging.

B.3. Geopolitical Blocs

Lastly, Figure 14 shows the difference in effects between two geoeconomic blocs: the western block and the rest of the world. The western bloc is again defined using the country's Ideal Point Distance (IPD). We use the sample median to recenter the state variable of the transition function so that the transition across blocs occurs around the median.

Public debt responses to a geoeconomic fragmentation shock differ across geopolitical blocs, with Western economies exhibiting a larger increase in public debt on impact relative to non-Western economies. This relative sensitivity is equivalent to roughly 2 percentage points of GDP for the Western bloc, with that difference becoming statistically insignificant after the initial impact and the debt dynamics converging at longer horizons.[6] These differences are consistent with the underlying macro-fiscal channels (Figure 15). Western economies experience a larger and more persistent output contraction, with real output falling by about 1 to 2 percentage points more than in non-Western economies. Revenue responses do not differ significantly across blocs, but Western economies display a stronger spending response in the first two years after the shock—by about 0.5 to 1 percentage point of GDP. As a result, fiscal balances deteriorate more in Western economies in the short run. Sovereign spreads show no significant differential response across blocs, suggesting that global financial forces dominate bloc-specific risk pricing. Inflation is temporarily lower in Western economies in the first two years, consistent with weaker demand conditions. Overall, the bloc-based results mirror the income-based

[6] Notice that this differential effect does not mean an absolute deterioration in public debt per se for each of the two blocs, as the bloc identification and the empirical strategy do not allow identifying the dynamics in the two groups separately.

heterogeneity: Western and richer economies experience deeper output losses and larger short-run debt increases, mainly through real-side adjustments and stronger fiscal responses.

B.4. Institutional quality and capital account openness

This section examines how the effects of fragmentation vary with structural characteristics, focusing on institutional quality and capital account openness. Institutional quality is measured using the World Bank's Worldwide Governance Indicators (WGI), which cover six governance dimensions: Voice and Accountability, Political Stability, Government Effectiveness, Regulatory Quality, Rule of Law, and Control of Corruption. Following common practice, we average these dimensions for each country-year to construct a composite index, with higher values indicating stronger institutions. Capital account openness is measured using the Chinn-Ito (KAOPEN) index, which summarizes restrictions on cross-border financial transactions using information from the IMF's AREAER. We use the original non-standardized index to preserve both cross-country and time-series variation, with higher values indicating greater financial openness.

The results indicate stronger debt responses in countries with higher institutional quality and greater financial openness (Figure 16). This pattern is consistent with the broader heterogeneity results, which show larger debt increases in advanced and Western economies. Countries with stronger institutions and deeper financial integration generally have better access to domestic and international capital markets, enabling them to smooth fragmentation shocks through higher borrowing.

B.5. Sensitivity checks to the heterogeneity analysis

To assess which mediating factor plays the largest independent role, we estimate a joint state-dependent specification that includes all five smooth-transition functions and their interactions with the fragmentation shock simultaneously. The results indicate that trade exposure to geopolitically distant import partners retains the clearest independent differential effect on debt dynamics. However, given the close correlation among trade exposure, financial openness, income, institutional quality, and bloc membership, we interpret these results cautiously (Figure 17).

To assess whether the results depend on how state dependence is modeled, we replace the smooth-transition function (STF) with either the underlying state variable or a dummy based on its median value—for example, for income classification, the dummy equals one when GDP per capita exceeds the World Bank high-income threshold. The alternative specifications yield broadly similar qualitative results (Figure 18): across the five state variables, differential debt responses are generally positive at short horizons and then fade toward zero, or become modestly negative, by year four. The estimates are somewhat smaller and less precise than in the smooth-transition specification, but the main patterns remain intact. In particular, the linear interaction supports the role of trade-partner exposure over the first three years, while the dummy specification confirms the early income-based differential. Overall, the direction and timing of the state-dependent effects are robust, even though their magnitude and statistical significance vary across interaction specifications.

IV. Debt Forecast Distribution (Debt-at-Risk)

To assess how geoeconomic fragmentation affects not only the average but also the distribution of future public debt outcomes, we adopt a location–scale quantile regression framework in line with the IMF's debt-at-risk methodology (Furceri and others, 2025). This approach allows us to decompose the effect of fragmentation

shocks into: (i) a location effect, capturing how shocks shift the center (*median*) of the future public debt distribution; and (ii) a scale effect, capturing how shocks affect the uncertainty about future debt.

A regression model is determined for each percentile of interest. Let $\tau_q(\cdot)$ denote the qth percentile of $F(y|\mathbf{x})$, i.e., $\tau_q(\mathbf{x}) = F_Y^{-1}(q) = inv\{y: F_Y(y|\mathbf{x}) \geq q\}$. Hence, to obtain the effect across different quantiles of the debt-to-GDP distribution at each horizon *h*, we estimate an additive regression model for panel data in which conditional percentiles are obtained as a function of the same covariates as above:

$$\tau_q\left(y_{i,t+h}\right) = X'\beta + (X'\pi)\,\tau_q(u_i) = \quad (5)$$
$$\gamma_i^{h,q} + a_0^{h,q} + \sum_{j=0}^{p}\beta_j^{h,q} shock_{t-j} + \sum_{j=1}^{p}\boldsymbol{c}_j^{h,q}\boldsymbol{Y}_{i,t-j} + \left(\delta_i^{h,q} + a_1^{h,q} + \sum_{j=0}^{p}\pi_j^{h,q} shock_{t-j} + \sum_{j=1}^{p}\boldsymbol{d}_j^{h,q}\boldsymbol{Y}_{i,t-j}\right)\tau_q(v_i),$$

where $\beta_0^{h,q}$ indicates the location effect; $\pi_0^{h,q}$ is the effect on uncertainty (scale) and $\beta_0^{h,q} + \pi_0^{h,q}\tau_q(u_i)$ is the total effect at the qth quantile of the debt distribution at horizon *h* of a 1 standard deviation shock in geopolitical fragmentation. This decomposition mirrors the IMF's growth-at-risk and debt-at-risk frameworks (Adrian, Boyarchenko, and Giannone, 2019; Furceri and others, 2025; respectively), where adverse shocks can simultaneously shift the center of the distribution and increase the probability of extreme outcomes.

Figure 19 reports the estimated location effects (left panel), scale effects (middle panel), and the implied impact on the 95th percentile of the public debt future distribution (right panel)—Furceri and others, (2025) measure of *debt-at-risk*.[7] The location IRFs confirm the baseline results for average public debt: fragmentation significantly raises the central tendency of the future debt distribution. The scale IRFs further indicate a significant widening of the distribution, with dispersion increasing by roughly 1 percentage point of GDP in the medium term. Together, these effects imply that fragmentation has a larger impact on the upper tail of future debt outcomes than on the lower tail, suggesting that debt realizations following fragmentation shocks may exceed those implied by average responses alone. The conditional 95th percentile of the projected debt distribution rises by almost 4 percentage points of GDP on impact, although statistical significance fades after about two years. This means that, relative to a no-shock counterfactual, fragmentation shocks raise debt-at-risk by nearly 4 percentage points of GDP in the short run. To provide an economic-magnitude benchmark for the Debt-at-Risk response, we use the estimated location-scale model to generate, for each country-year observation in the estimation sample, the fitted conditional 95th percentile of debt implied by the model. We then compute the annual within-country change in this fitted upper-tail debt measure and calculate its standard deviation across the sample, which equals 5.3 percentage points of GDP. The estimated effect of a one-standard-deviation fragmentation shock on the conditional 95th percentile is expressed relative to this historical variation. This benchmark indicates how large the estimated tail response is compared with typical annual movements in the model-implied upper-tail debt measure. A one-standard-deviation fragmentation shock raises the conditional 95th percentile of future debt by 4 percentage points of GDP, equivalent to approximately 0.75 standard deviations (4/5.3) of the historical annual variation in the model-implied upper-tail debt measure.

Fragmentation also increases the scale of the distribution of future GDP growth, widening downside risks (Figure 20).[8] Beyond lowering expected future growth, fragmentation also widens the growth distribution and

[7] For other outcomes no significant effects for the scale parameter were found, so they are not shown here, but remain available upon request.

[8] The location-effect IRF for debt coincides with the corresponding linear LP IRF reported in Figure 2 (top-center panel). This follows from the structure of the location-scale LP estimator: under the normalization that the standardized disturbance has median zero, the conditional median is equal to the location component. Because the location equation is specified using the same linear LP specification as in the baseline analysis, its estimated IRF coincides with the baseline LP response. For log GDP, the

worsens downside risks. The positive scale effect in Figure 20 indicates greater dispersion, while the decline in the 5th percentile shows a deterioration in growth-at-risk. Overall, fragmentation shocks shift the lower tail of future growth downward, increasing the likelihood of weak or negative growth outcomes.

Overall, the results show that geoeconomic fragmentation affects not only average debt and growth outcomes, but also the risks around them. By widening the conditional distributions of public debt and growth, fragmentation shocks amplify macro-financial uncertainty and increase the likelihood of adverse tail events, even when average effects appear moderate.

V. Conclusions

This paper shows that geoeconomic fragmentation has persistent macro-fiscal costs. Using a new Geopolitical Fragmentation Index and a two-stage strategy combining panel SVARs and local projections, we find that fragmentation shocks lower output, widen fiscal deficits, and raise public debt by about 1 to 3 percentage points of GDP over the medium term. These effects reflect weaker activity, higher spending, lower revenues, and a temporary tightening of financing conditions, while also raising *debt-at-risk* and worsening *growth-at-risk*.

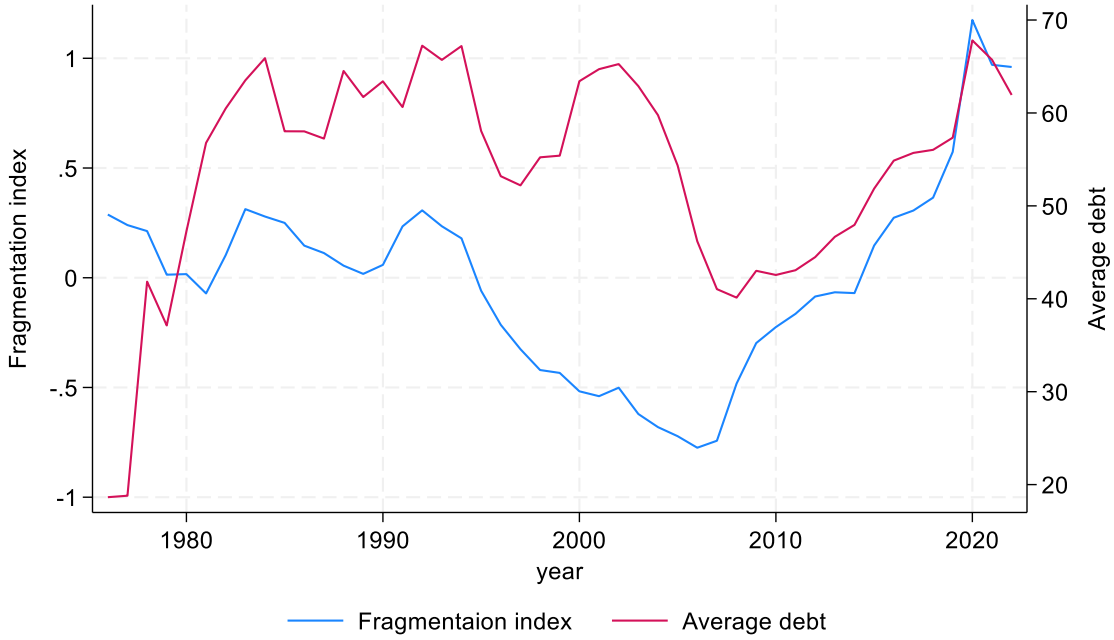
These findings imply that fragmentation is both a source of fiscal pressure and a driver of macro-financial risk. Fiscal frameworks should therefore account more explicitly for geopolitical risks through prudent debt anchors, stress scenarios in debt sustainability analysis, and careful design of fiscal responses. Because vulnerability depends more on structural exposures than on broad country groups, countries highly reliant on geopolitically distant trade partners may benefit from supplier diversification, friend-shoring, and more resilient strategic supply chains.

Future research could examine how fiscal rules, fiscal councils, and domestic political factors shape the debt effects of fragmentation shocks.

corresponding linear LP IRF is not reported in the Results subsection, where Figure 3 presents only the SVAR estimates, but it likewise coincides with the location-effect IRF shown here and is available from the authors upon request.

Figures and Tables

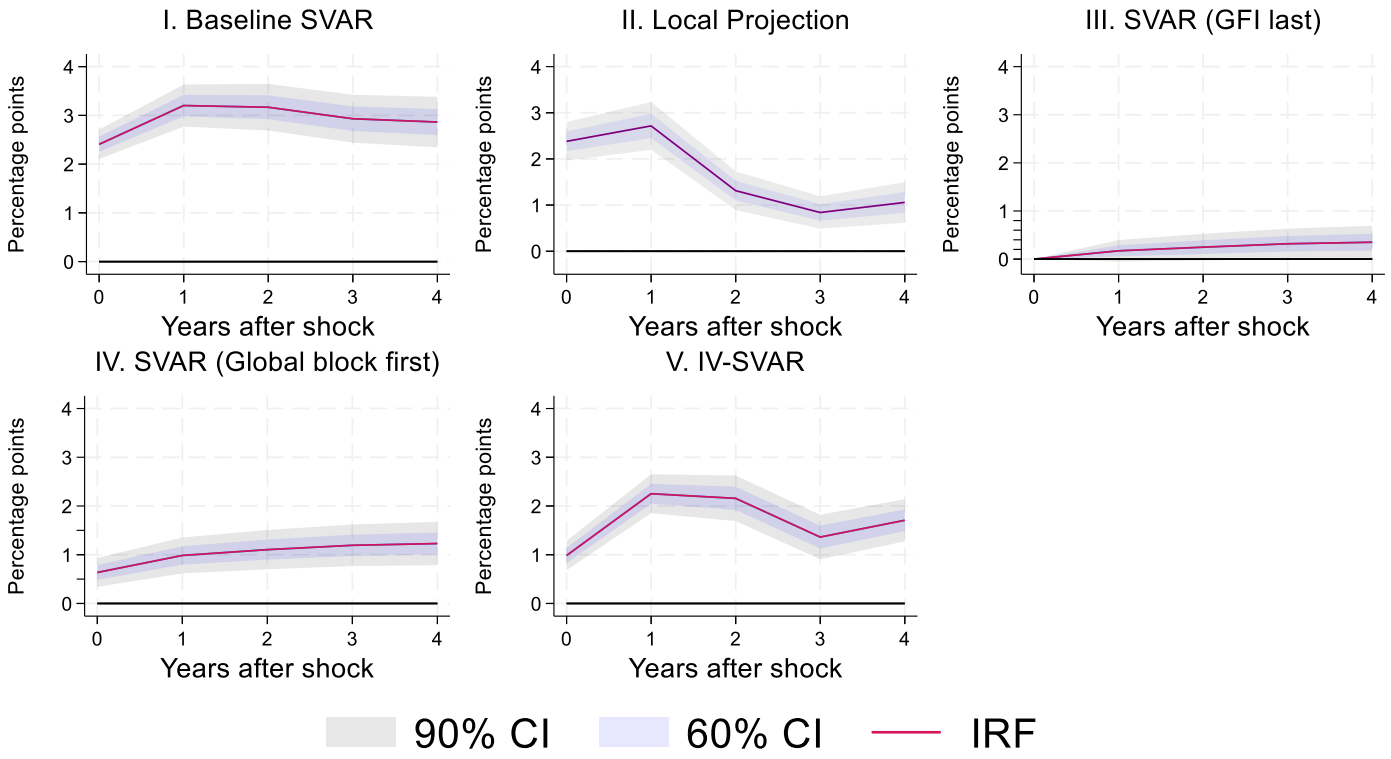
Figure 1. Geoeconomic Fragmentation and Average Global Debt



Sources: Fernández-Villaverde, Mineyama, and Song (2024); and World Economic Outlook (WEO) database.

Notes: The global average is the sample average of the debt-to-GDP ratio over all countries with data available in the WEO database in a given year. The figure is based on the two raw time series for geoeconomic fragmentation and average global public debt starting its coverage from 1976 until 2022.

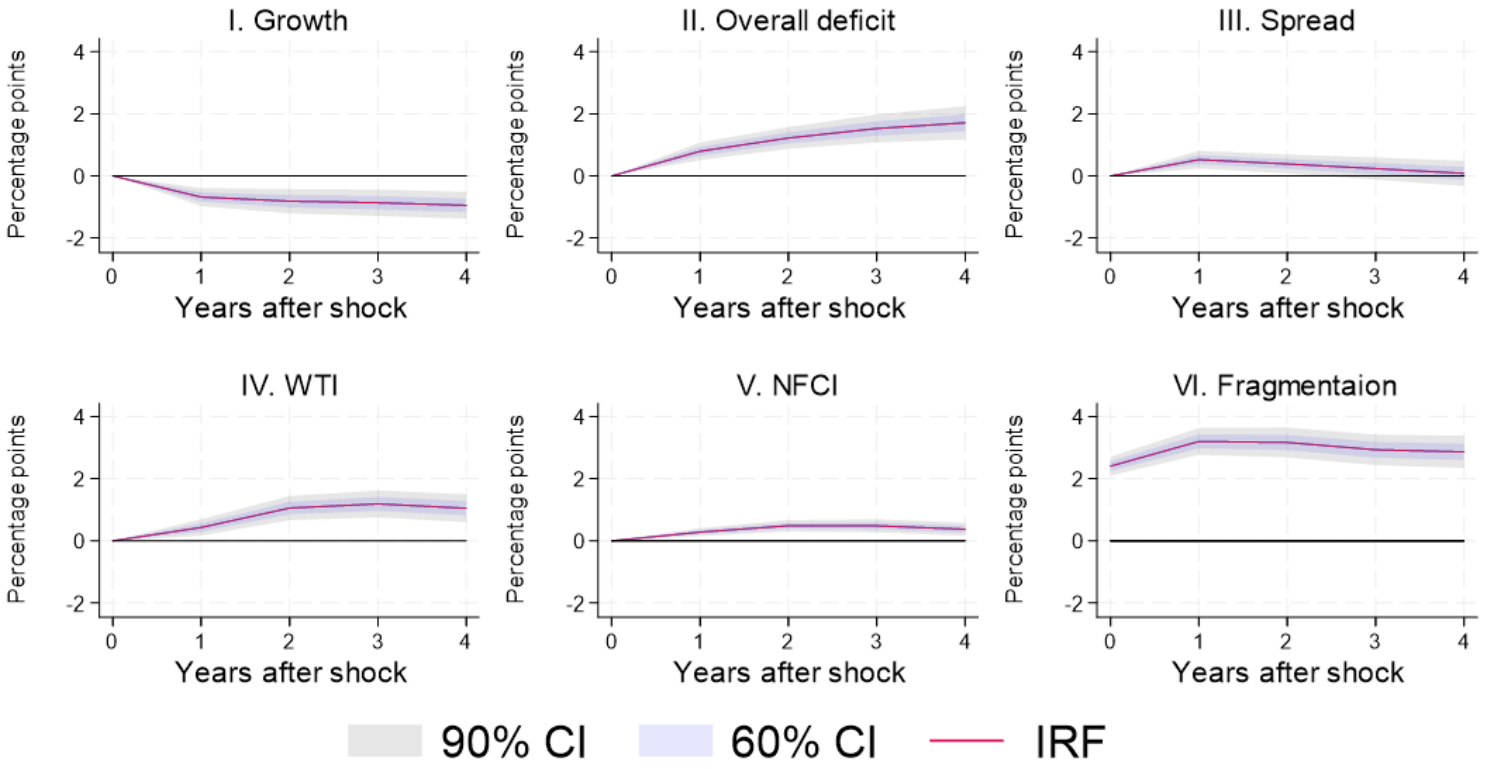
Figure 2. Fragmentation Effects on Public Debt by Alternative Identification Schemes, 1990–2022



Sources: Authors' estimations.

Notes: The purple line indicates the Impulse Response Function to a 1 standard deviation fragmentation shock. The grey (light blue) area 90 (60) percent confidence intervals (CI) based on standard errors clustered by country. The charts shows estimates from the following identification schemes: recursive SVAR with the baseline ordering (the top-left); local projection (LP) estimates based on Equation (3) (top-middle); recursive SVAR with the GFI ordered last (top-right); an intermediate recursive SVAR in which the global block precedes the GFI and the macro-fiscal variables (bottom-left), and a IV-SVAR (bottom center).

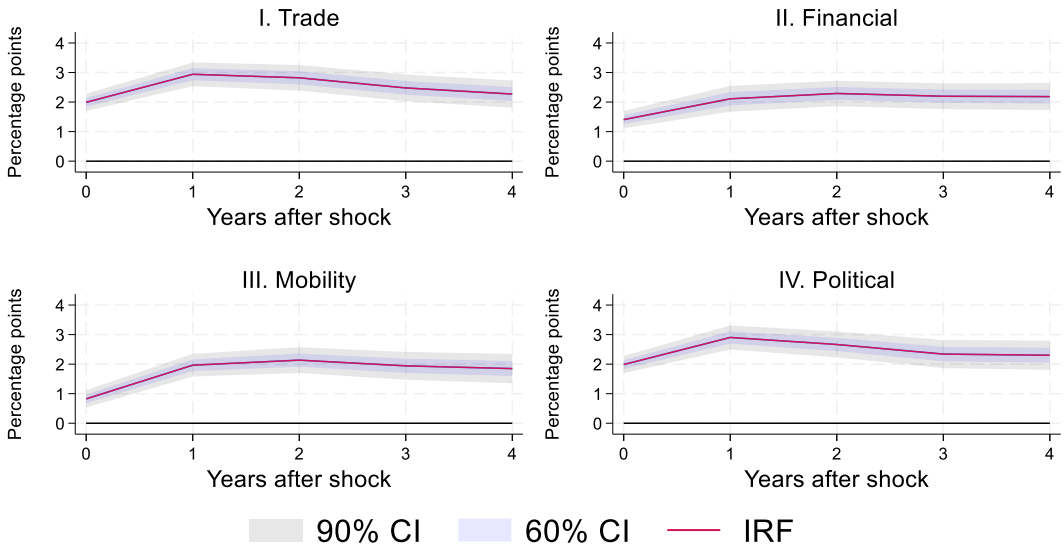
Figure 3. Effects of Other Shocks on Public Debt, 1990–2022



Sources: Authors' estimations.

Notes: The purple line indicates the Impulse Response Function to a 1 standard deviation shock to each outcome. The grey (light blue) area 90 (60) percent confidence intervals (CI) based on standard errors clustered by country. The charts show estimates of the debt IRF to a 1 standard deviation shock to growth, overall deficit, sovereign bond spread, the crude oil price (WTI) index, the Chicago Fed National Financial Conditions Index (NFCI), and geoeconomic fragmentation itself—all from the baseline identification scheme.

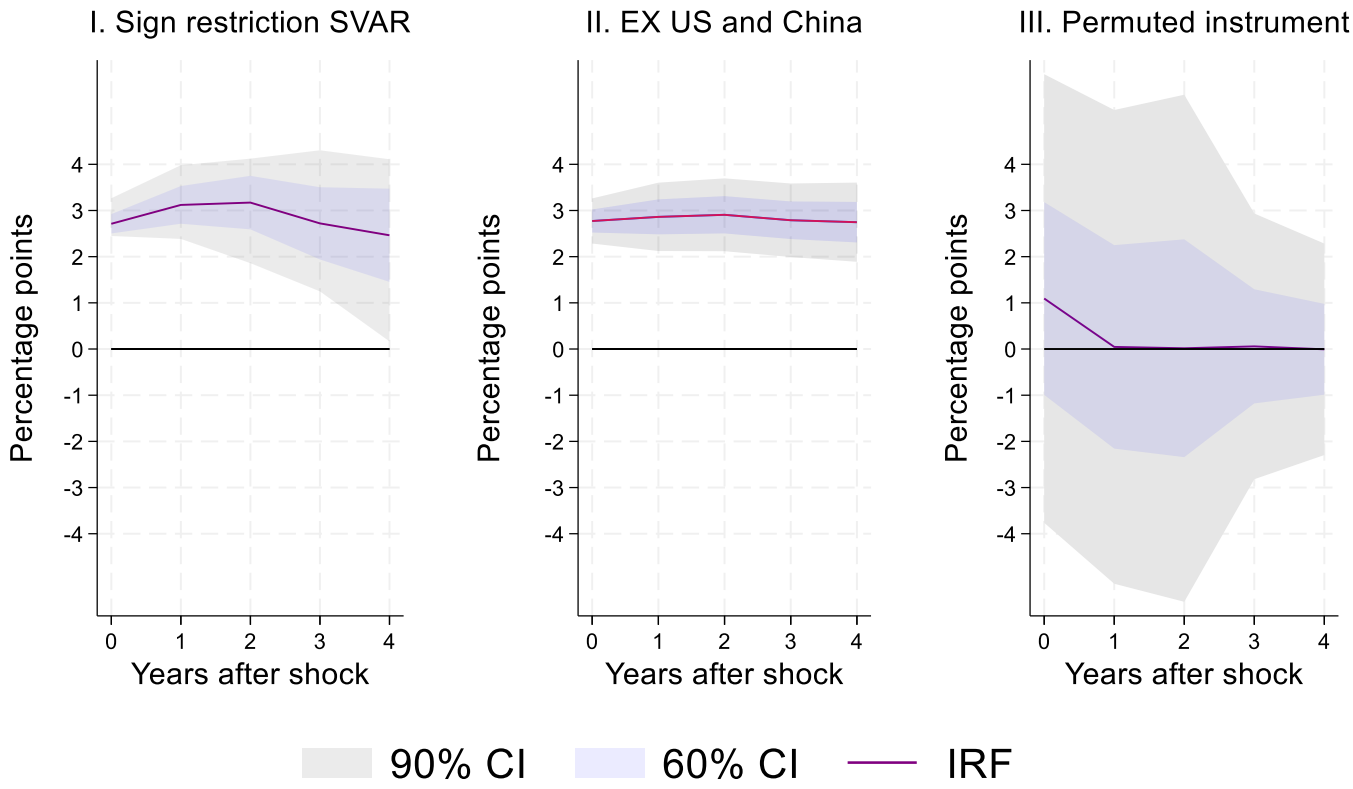
Figure 4. Effects of Subcomponents of Fragmentation on Public Debt, 1990–2022



Sources: Authors' estimations.

Notes: The purple line indicates the Impulse Response Function to a 1 standard deviation fragmentation shock. The grey (light blue) area 90 (60) percent confidence intervals (CI) based on standard errors clustered by country. The charts show debt IRF estimates from the baseline identification schemes for each subcomponent of the geoeconomic fragmentation index.

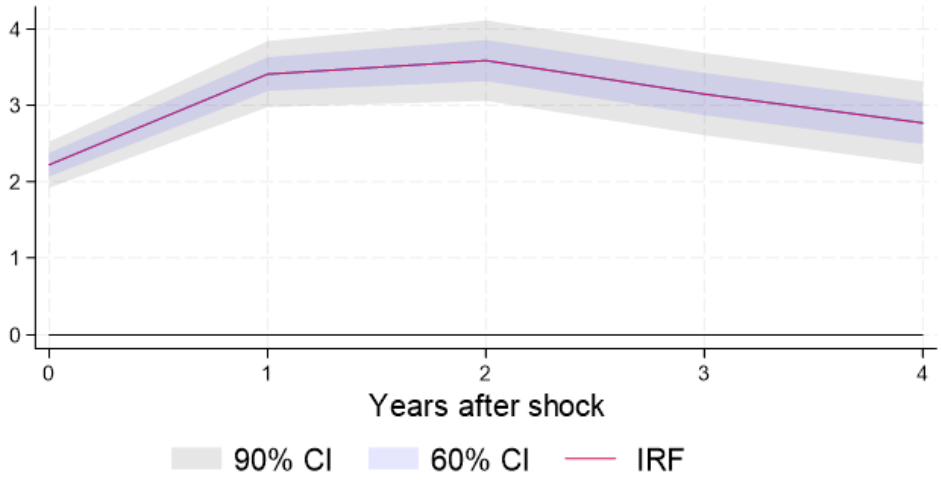
Figure 5. Robustness Checks for Debt-on-GDP, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals (CI) based on standard errors clustered by country. The charts show the debt IRF from three different sensitivity checks: sign-restricted Bayesian SVAR with a mean group averaging; baseline SVAR on a restricted sample that excludes China and the US; IV-SVAR in which the dates of the geoeconomic events in the original instrument are randomly permuted.

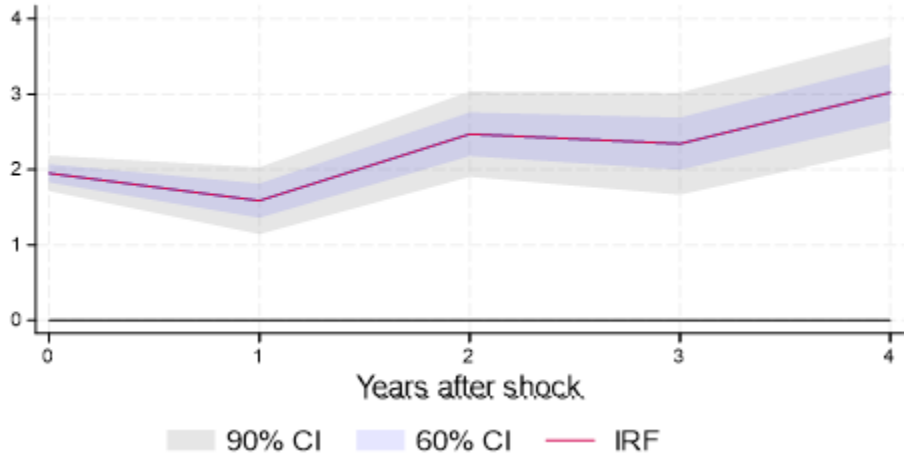
Figure 6. IV-SVAR Debt Effects with a Modified Instrument, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals (CI) based on standard errors clustered by country. The IRF is obtained from an IV-SVAR that uses the original narrative instrument stripped by those events that most likely affect fiscal aggregates.

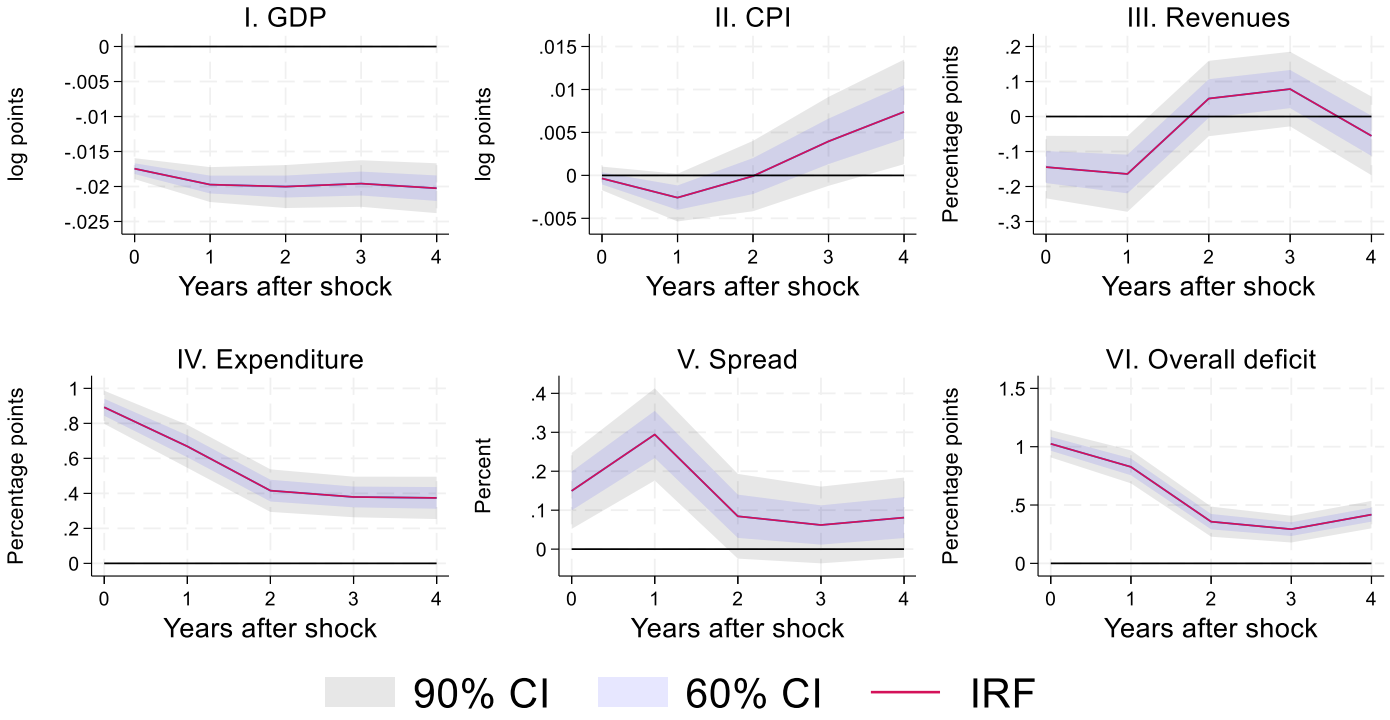
Figure 7. Adding Special Controls to the Baseline SVAR 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals (CI) based on standard errors clustered by country. The chart shows the debt IRF obtained with the baseline SVAR adding COVID19 (2020-21) and Global Financial Crisis (2008-09) dummies as well as the Caldara and Iacoviello Geopolitical Risk Index as controls.

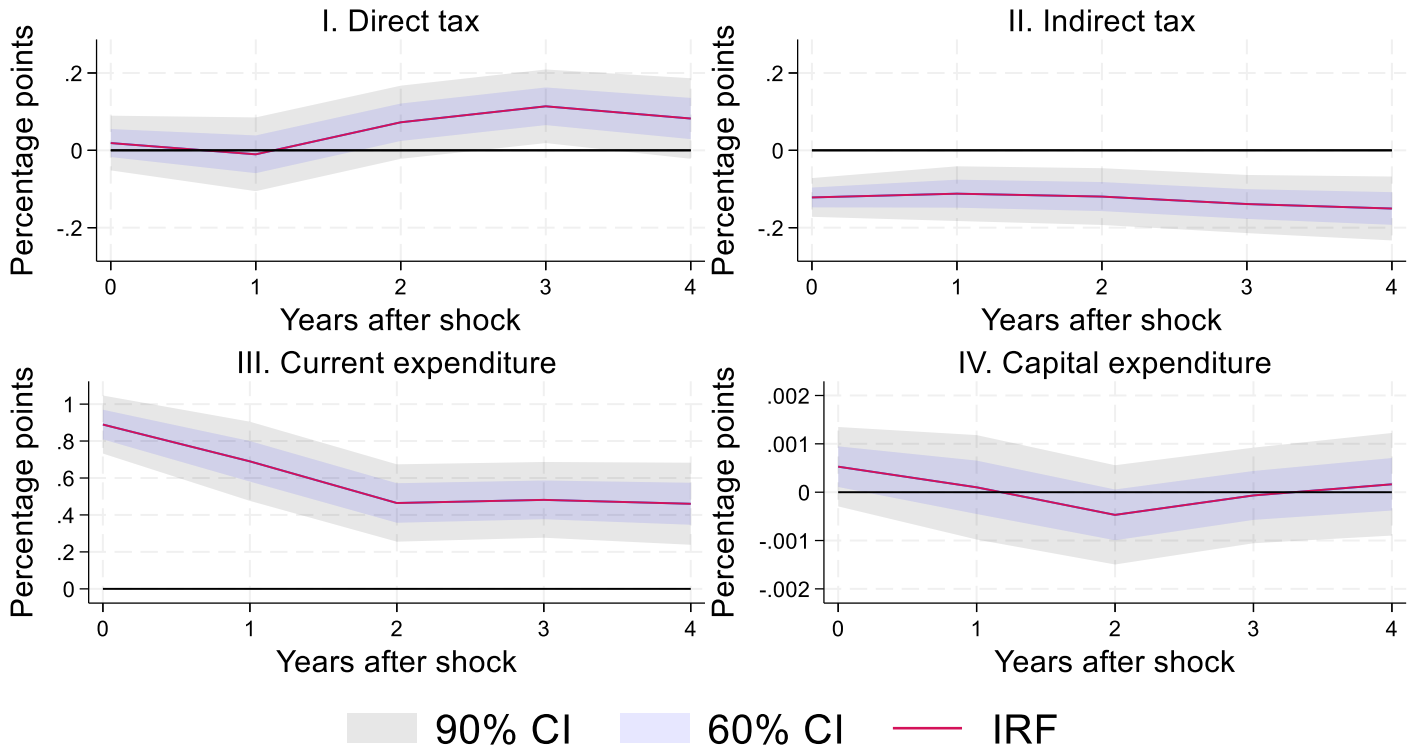
Figure 8. Baseline SVAR Estimates of the Macroeconomic Drivers of the Debt Effects of Fragmentation, 1990–2022



Source: Authors' estimation.

Note: The purple line indicates the IRF to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals (CI) based on standard errors clustered by country. The results on the overall deficit are obtained from the SVAR with the baseline ordering, where expenditure and revenues are replaced with the overall deficit.

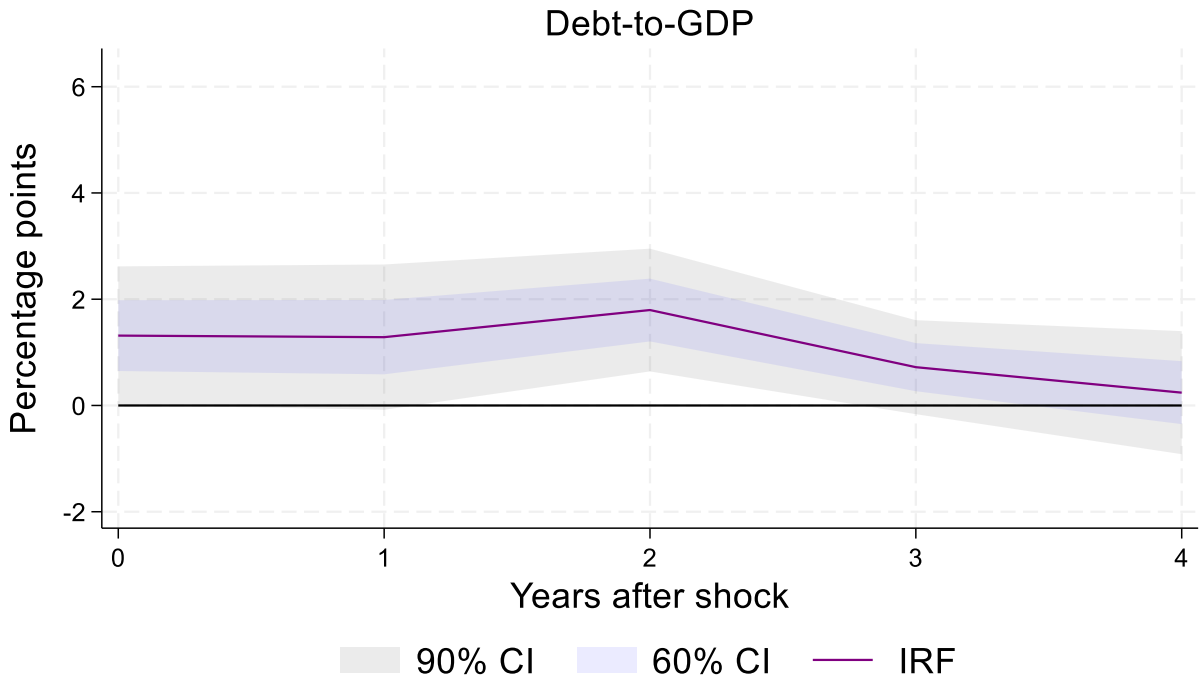
Figure 9. Effects of Geoeconomic Fragmentation on Public Revenues and Spending Subcomponents, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals (CI) based on standard errors clustered by country. Estimates are obtained using the SVAR with recursive identification by replacing the total spending and revenues with the corresponding subcomponents.

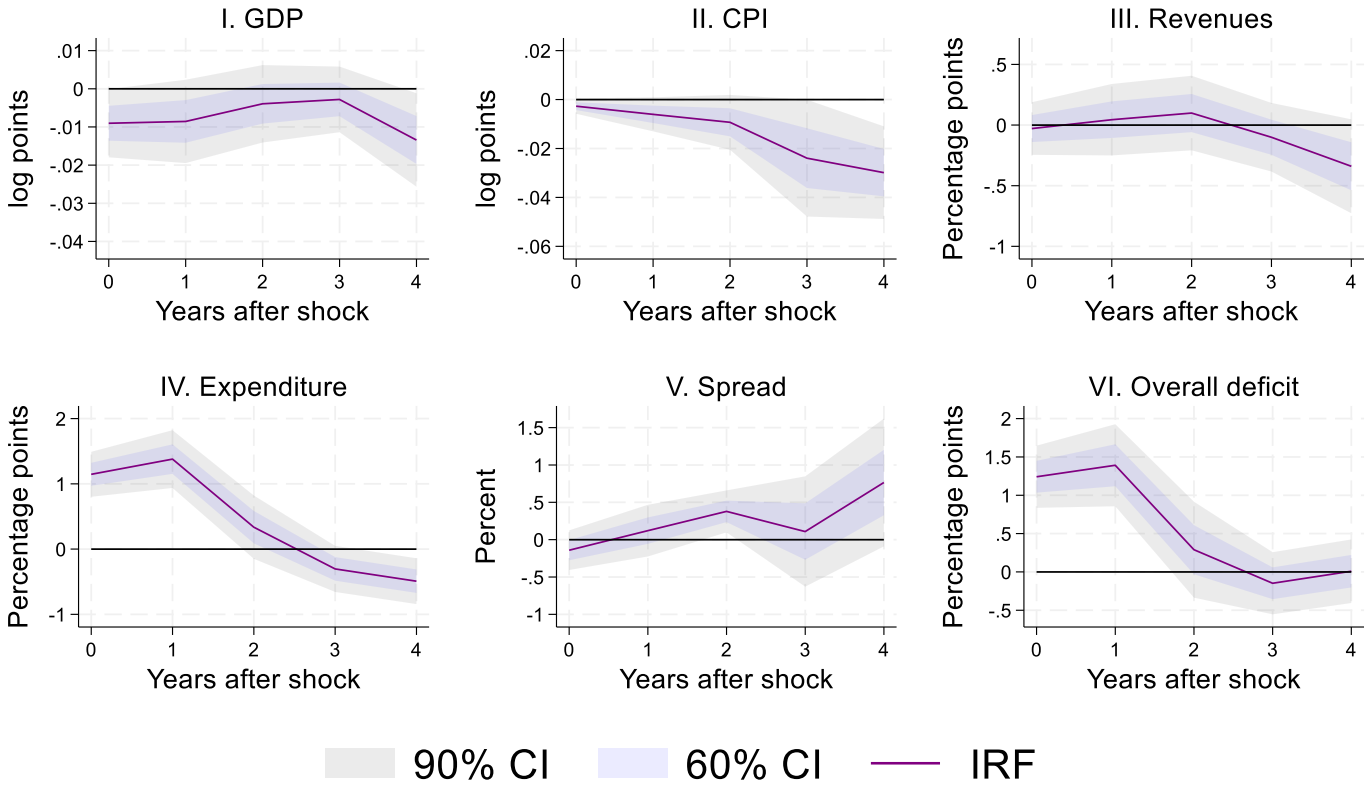
Figure 10. Differential Effects by Income (GDP per Capita) Level, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country. The state variable in this analysis uses GDP per capita to split the sample between high-income economies and low-middle income economies. The IRFs indicate the differential effect in high- *vis a vis* low-income economies.

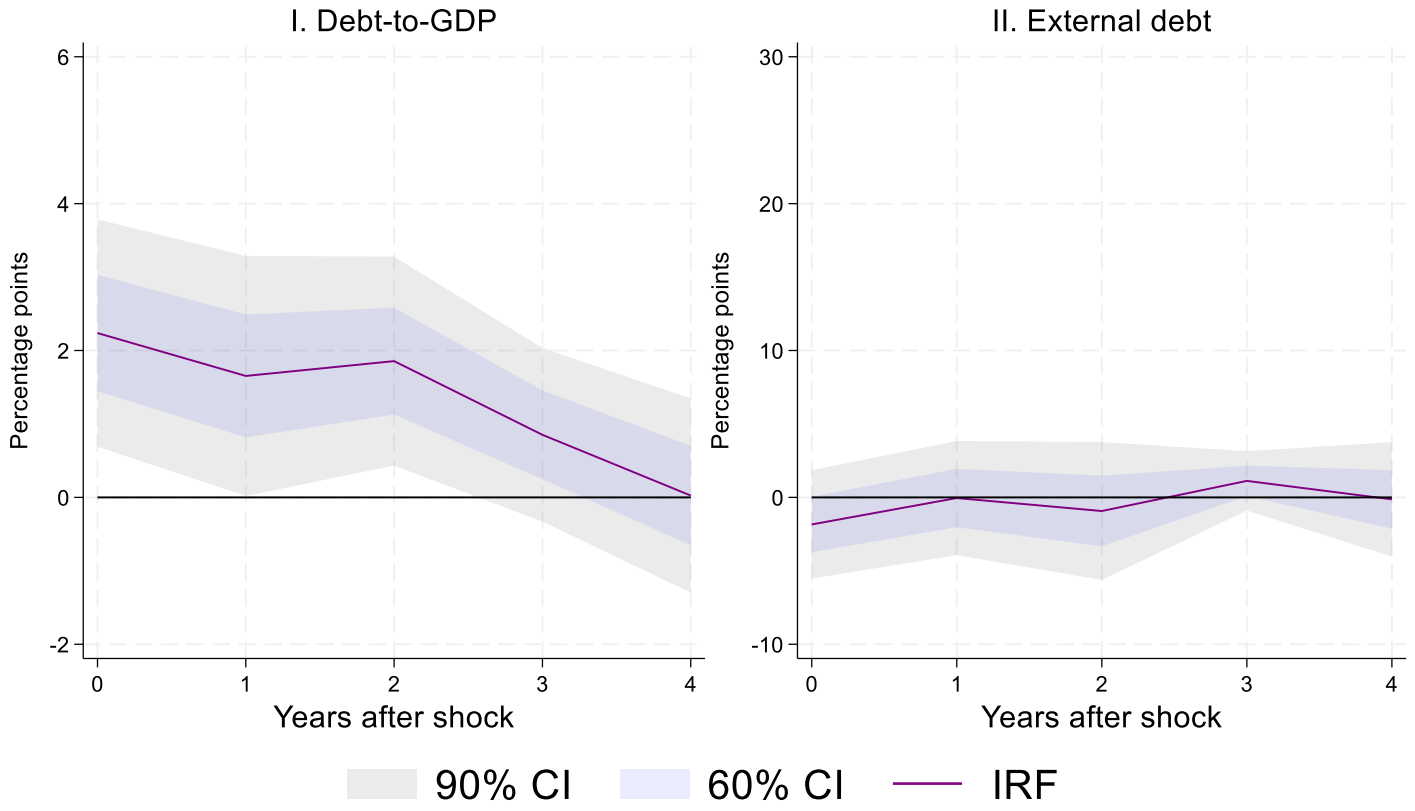
Figure 11. Differential Effects by Income Level: Macro-Fiscal Transmission Channels, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country. The state variable in this analysis uses GDP per capita to split the sample between high-income economies and low-middle income economies. The IRFs indicate the differential effect in high- *vis a vis* low-income economy.

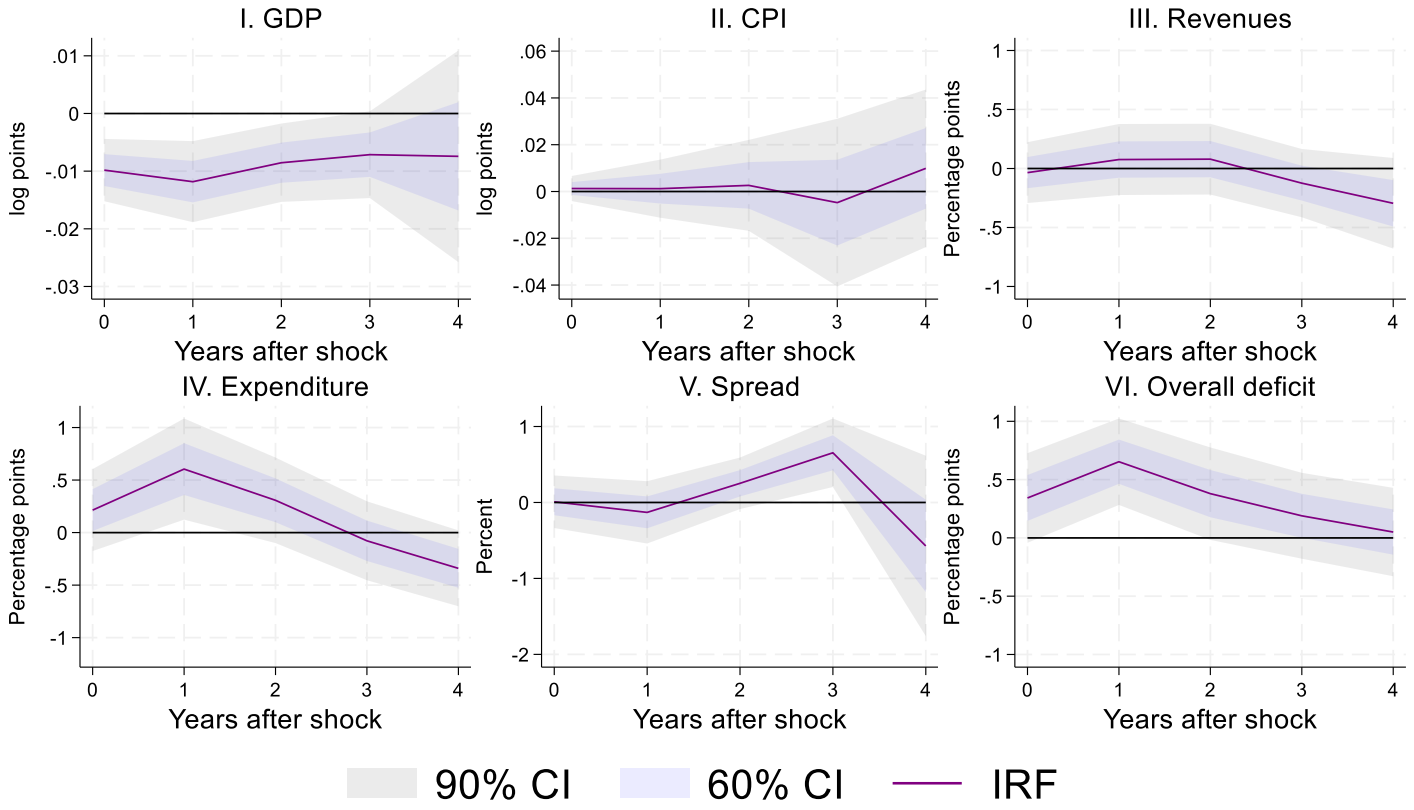
Figure 12. Effects by Level of Trade Exposure to Geopolitical Distant Partners, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country. The results on the overall deficit are obtained from a SVAR where expenditure and revenues are replaced with the overall deficit. Trade exposure to geopolitical distant partners uses an index of import dependence on geopolitical distance, measuring the extent to which a country relies on imports from trade partners that are ideologically distant from itself. Positive values in the charts indicate a higher effect of geofragmentation on public debt for countries more exposed to trade with geopolitical distant partners. The IRFs indicate the differential effect in high- vis a vis low-exposure to import dependence from geopolitically distant trade partner.

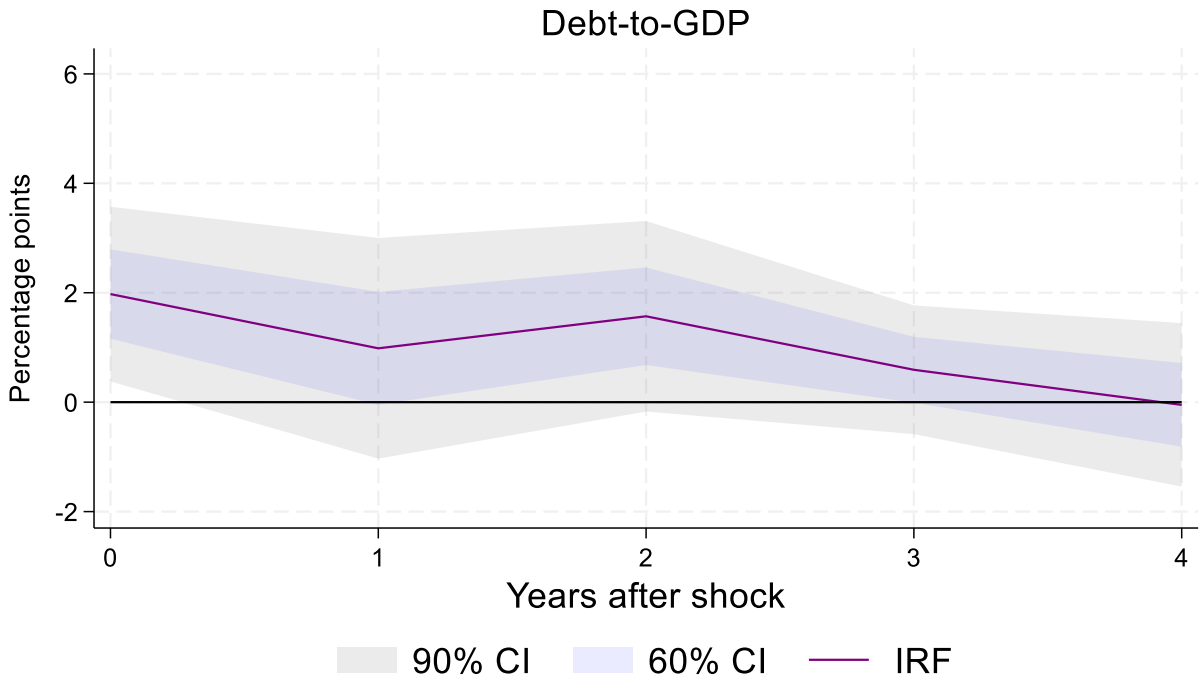
Figure 13. Differential Effects by Exposure to Geopolitical Distant Partners: Macro-Fiscal Transmission Channels, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country. The results on the overall deficit are obtained from a SVAR where expenditure and revenues are replaced with the overall deficit. Trade exposure to geopolitical distant partners uses an index of import dependence on geopolitical distance, measuring the extent to which a country relies on imports from trade partners that are ideologically distant from itself. Positive values in the charts indicate a higher effect of geofragmentation on public debt for countries more exposed to trade with geopolitical distant partners. The IRFs indicate the differential effect in high- vis a vis low-exposure to import dependence from geopolitically distant trade partner.

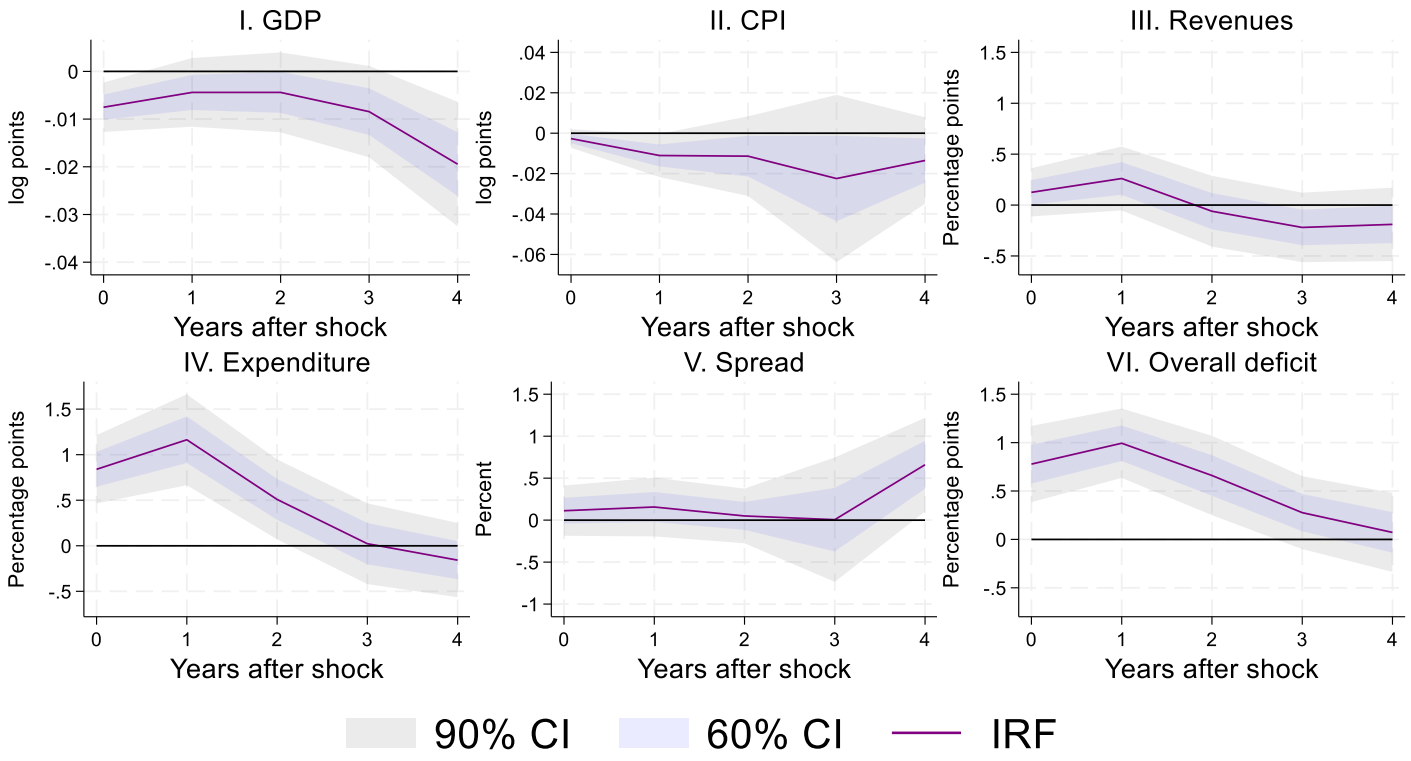
Figure 14. Differential Effects by Geoeconomic Bloc, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country. The IRF indicates the differential effect in west-aligned countries vis a vis the Rest of the World. West-aligned countries are those with an above-median value of the country's Ideal Point Distance (IPD). See Section II.B.2 for the IPD's definition.

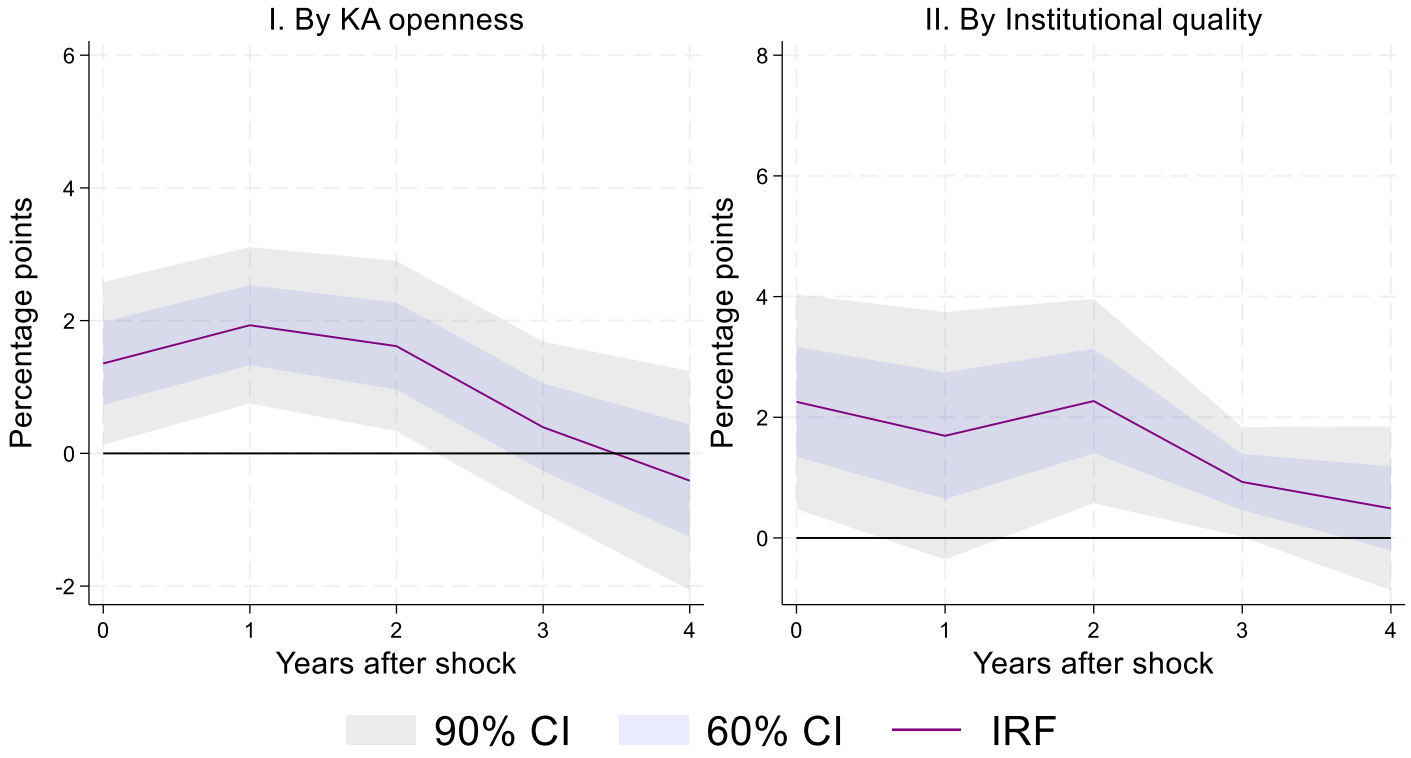
Figure 15. Differential Effects by Geoeconomic Block: Macro-Fiscal Transmission Channels, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country. The results on the overall deficit are obtained from an SVAR where expenditure and revenues are replaced with the overall deficit. The IRFs indicate the differential effect in west-aligned countries vis a vis the Rest of the World. West-aligned countries are those with an above-median value of the country's Ideal Point Distance (IPD). See Section II.B.2 for the IPD's definition.

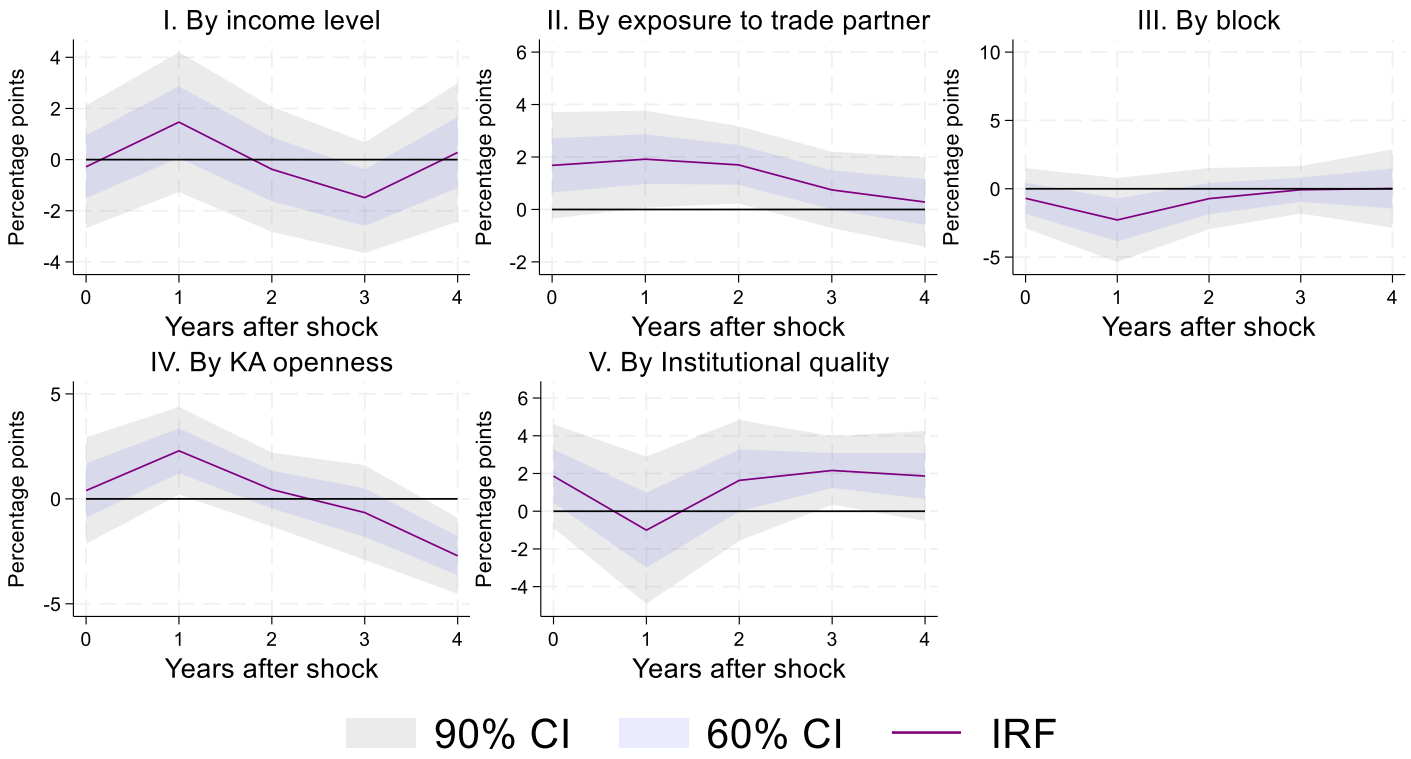
Figure 16. Differential Effects by Capital Account Openness and Institutional Quality, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country.

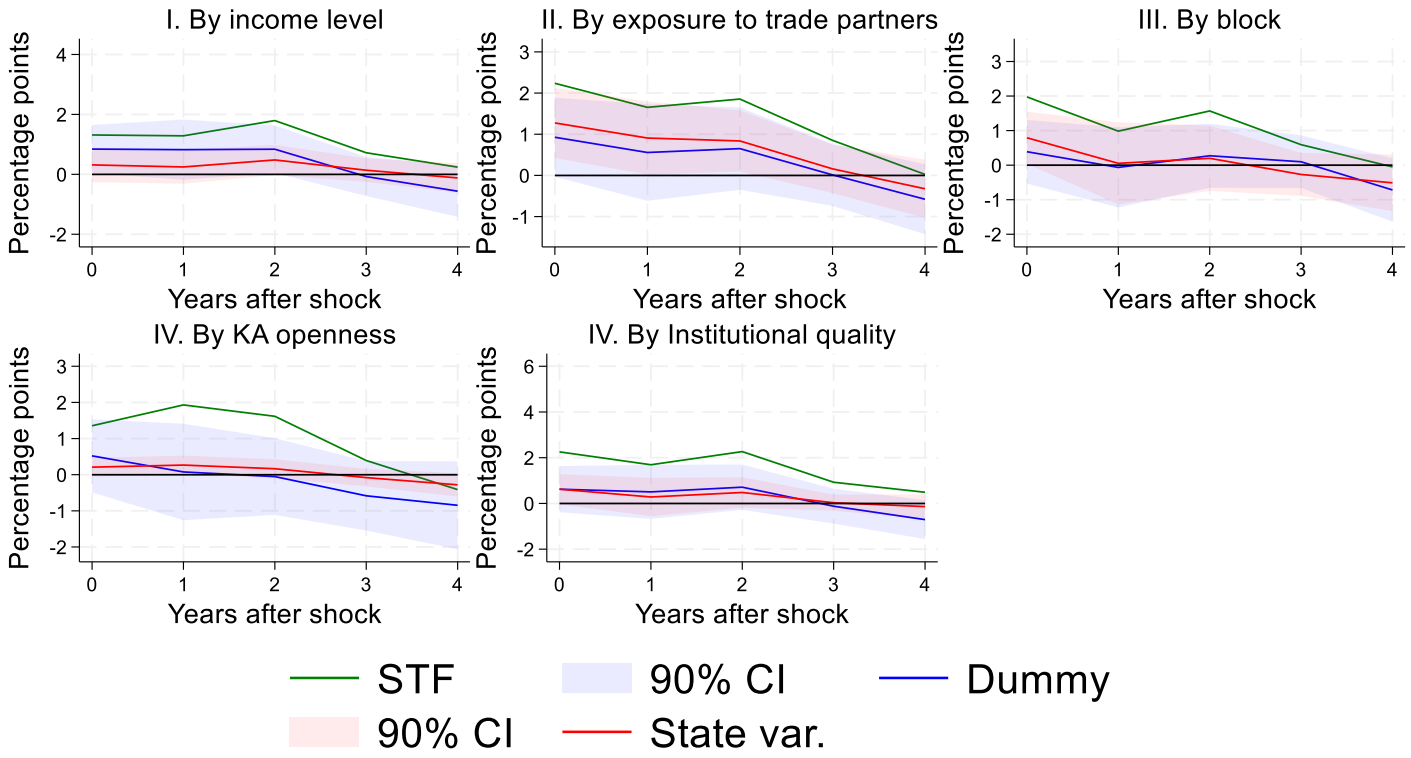
Figure 17. Differential Effects on Debt Including All Interactions, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country. The charts show the debt IRF when all interactions are included at the same time.

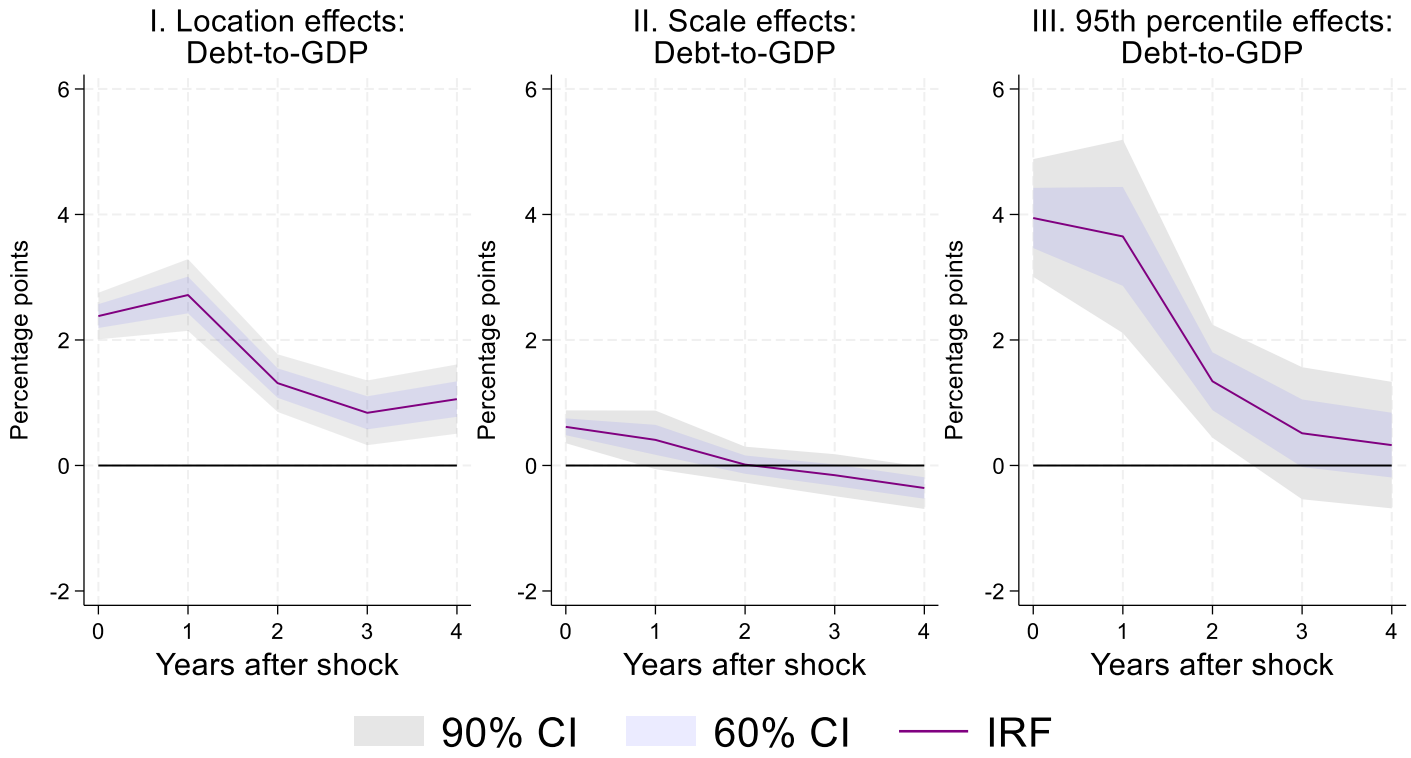
Figure 18. Differential Effects by Different Specification of State-dependency, 1990–2022



Source: Authors' estimations.

Note: The green line indicates the IRF from the original smooth-transition function specification, for which no confidence bands are reported. The blue (red) line indicates the IRF when a dummy (the underlying state variable) is used instead of the smooth transition function (STF) in the interactions. Confidence bands are reported in light blue (red) for these two alternative specifications.

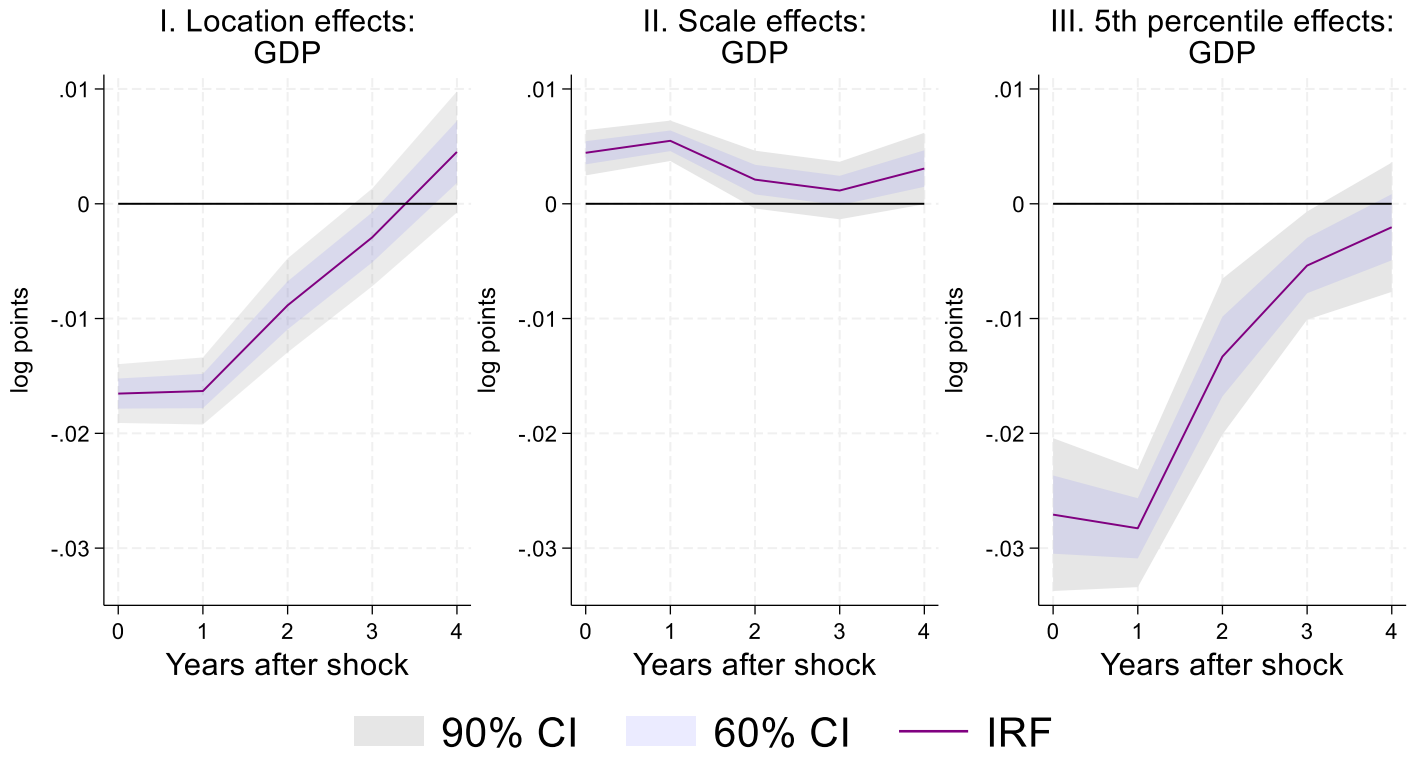
Figure 19. Estimated Effects on Location, Scale, and *Debt-at-Risk* of the Projected Debt Distribution, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country. The location effect captures how shocks shift the center (median) of the distribution of future debt. The scale effect captures how shocks affect the dispersion of the distribution of future debt. The 95th percentile effect captures how shocks shift the 95th percent (upper tail) of the distribution of future debt—Furceri and others (2025) measure of *debt-at-risk*.

Figure 20. Estimated Effects on Location, Scale, and *Growth-at-Risk* of the Projected Growth Distribution, 1990–2022



Source: Authors' estimations.

Note: The purple line indicates Impulse Response Functions (IRFs) to a 1 standard deviation fragmentation shock. The grey (light blue) area corresponds to 90 (60) percent confidence intervals based on standard errors clustered by country. The location effect captures how shocks shift the center (median) of the distribution of future economic growth. The scale effect captures how shocks affect the dispersion of the distribution of future growth. The 5th percentile effect captures how shocks shift the 5th percent (lower tail) of the distribution of future growth— Adrian, Boyarchenko, Giannone's (2019) measure of *growth-at-risk*.

Table 1. Sign Restrictions of Each Shock (Rows) on a Given Variable

	GFI	s	gdp	ob	d
Fragmentation shock	+	+	-	-	+
Fiscal austerity shock			-	+	-

Source: Authors' analysis.

Appendix

Table A1: Descriptive statistics

	Mean	St. Dev.	Min	Max	Obs.
GFI	-0.069	0.544	-0.775	1.174	2,515
Spread	2.676	5.897	-30.374	98.027	2,515
Overall balance	2.019	5.393	-36.14	37.726	2,515
log GDP	6.938	3.052	-2.596	16.577	2,515
Debt to GDP	56.302	38.102	0	357.687	2,515
log global GDP	17.062	0.31	16.398	17.509	2,515
log WTI	4.77	0.547	3.503	5.439	2,515
log SP500	7.335	0.524	6.03	8.36	2,515
VIX	19.893	6.1	11.09	32.693	2,515
NFCI	-0.382	0.445	-1.02	1.219	2,515

Source: Authors' calculations.

Note: The statistics are retrieved from the sample used in the baseline estimation. Obs. = Observations.

Table A2: List of countries in the estimation sample by income level

High-income economies	*Low- and Middle-income Economies*
Australia	Algeria
Austria	Angola
Bahrain	Argentina
Barbados	Bosnia and Herzegovina
Belgium	Botswana
Brunei Darussalam	Brazil
Canada	Chad
Chile	China
Croatia	Congo, Republic of
Cyprus	Egypt
Czech Republic	Equatorial Guinea
Denmark	Eswatini
Finland	Gabon
France	Georgia
Germany	Ghana
Greece	Guinea
Guyana	Haiti
Hungary	Indonesia
Iceland	Iran
Ireland	Jordan
Israel	Kazakhstan
Italy	Kyrgyz Republic
Japan	Lebanon
Korea	Lesotho
Kuwait	Malawi
Luxembourg	Malaysia
Malta	Mauritius
Netherlands	Mexico
New Zealand	Moldova
Norway	Morocco
Oman	Mozambique
Panama	Nicaragua
Poland	Niger
Portugal	Pakistan
Russia	Paraguay
Saudi Arabia	Philippines
Singapore	Sierra Leone
Slovak Republic	South Africa
Slovenia	Suriname
Spain	Tajikistan
Sweden	Tanzania
Switzerland	Thailand
Taiwan Province of China	Ukraine
United Kingdom	Yemen
United States	Zambia

Table A3: Narrative Events of the Instrument

(A) Surprise war outbreaks, international conflicts, terrorism
Iraqi invasion of Kuwait*(1990:Q3), Gulf War*(1991:Q1), NATO intervention in Serbia*(1994:Q2) and Kosovo*(1999:Q1), 9.11*(2001:Q3), Iraq War*(2003:Q1), U.S. strike on ISIL*(2014:Q3), Russia's invasion of Ukraine*(2022:Q1)
(B) Unforeseen geopolitical shifts
Geneva Accords between Afghanistan and Pakistan†(1988:Q2), Fall of the Berlin Wall†(1989:Q4), USSR dissolution events†(1988-1991), U.S.-Vietnam normalization†(1995:Q3), Arab Spring*(2010:Q4), Brexit vote*(2016:Q2)
(C) Enactment of trade deals, currency unions, or trade restrictions
NAFTA†(1994:Q1), WTO†(1995:Q1), Mercosur†(1995:Q1), Euro†(1999:Q1) and its expansion†(2004:Q2), U.S.-China trade war*(2018-2019)

Source: Fernández Villaverde, Mineyama, and Song (2024).
Note: There are 24 narrative episodes that make up the quarterly instrumental variable, with geopolitical fragmentation events (integration events) marked with an asterisk (dagger).

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